Quality, Reputation, and Reimbursement Design in the Hospice Industry

Rubaiyat Alam*

Kansas State University

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Abstract

I study hospice competition and explore the impact of alternative hospice reimbursement schemes by Medicare. Hospices can build up their reputation by consistently choosing high quality, which is costly to provide. They are paid a fixed per-patient per-day rate by Medicare. I develop and estimate a dynamic oligopoly model of the California hospice industry to first show hospices compete on reputation, and then recover the hospice cost function. Counterfactuals show that i) higher reputation persistence leads to improved hospice quality, ii) lower Medicare reimbursement worsens hospice quality, and iii) partially tying Medicare reimbursement to quality lowers Medicare spending without sacrificing equilibrium quality.

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1 Introduction

Hospices provide palliative care to dying patients. Several policy concerns about the hospice industry motivate this paper. First, hospice care is a low-cost alternative that can significantly reduce end-of-life spending by moving many patients away from painful ineffective care. However, we need to incentivize good hospice quality for patients to switch; in turn, this requires us to understand how hospices choose quality of service.[1] Second, hospices have complained about low Medicare reimbursement rates. While this has reduced Medicare spending, is there any effect on hospice quality? Finally, Medicare reimburses hospices for each day a patient is enrolled. Can we improve upon this reimbursement scheme?

Using a dynamic oligopoly model of the California hospice industry, I show that hospices dynamically choose quality to build up reputation and attract consumers. This involves estimating consumer demand for hospices as well as the hospice cost function. Then, I use my model and estimates to evaluate counterfactual policies. Making reputation more persistent – such as having a hospice review website – improves hospice quality. Reducing Medicare reimbursement of hospices worsens hospice quality. Finally, partially tying reimbursement to quality lowers Medicare spending without worsening hospice quality.

The hospice industry is useful to study for several reasons. First, the US hospice industry is large and growing – over 1.6 million Medicare patients used hospices in 2022, and Medicare spent around $21 billion on hospice services in 2019. However, very few papers empirically study incentives and regulation design for this industry. Second, it sheds light on policy design for "regulated price" healthcare settings where the government fixes reimbursement rates. Finally, it is a clean setting for studying how firm reputation affects consumer and firm choices, particularly in healthcare markets.

Hospice care is given to dying patients at their residences, where hospice staff administer pain medication and help with living arrangements. More visits by the hospice staff can be thought of

[1] Concerns have been raised about hospice quality across the US. The Department of Health and Human Services has released a report citing numerous deficiencies in hospice quality: https://oig.hhs.gov/newsroom/media-materials/media-materials-2019-hospice/

as implying higher quality – it means the hospice is checking up on patients more and adjusting for day-to-day difficulties. Therefore, my measure of hospice quality is the average visits-per-patient-day. There is no price competition because most hospice patients are covered by Medicare, and Medicare pays hospices a fixed per-day rate for each patient. This makes hospice reputation particularly salient for consumers. Hospices which have given higher quality care in the past are more likely to be known and referred within the community. This suggests that a hospice can build up its reputation by persistently choosing high quality over time, and use this improved reputation to attract more patients. I provide descriptive evidence to confirm my claim about visits-per-patient-day being an appropriate measure of quality and effort, as well as evidence suggesting that reputation is important for hospice choice by consumers.

To study reputation and quality choice in this setting, I build a structural model of hospice demand by consumers and dynamic quality choice by hospices. Reputation of a hospice is defined to be a stock of its current and past quality choices. This reputation stock partially depreciates when moving from one period to next. On the demand side, each consumer makes a discrete choice from a set of hospices in her market. Her choice is influenced by hospices' reputations and characteristics. On the supply side, hospices play a dynamic game in an oligopoly setting where they choose quality every period. They take into account that higher quality will lead to increased reputation and market share but also higher cost per consumer.

I estimate this model using annual hospice-level data from 39 counties in California for 2002-2018. The estimation proceeds as follows. First, I estimate a nested-logit discrete choice demand model following Berry (1994) and allowing for persistent unobserved demand shocks. This lets me pin down the magnitude and persistence of hospice reputation for consumer choice. Second, I estimate a dynamic oligopoly model of quality choice which incorporates reputation effects, following the method of Bajari et al. (2007). By incorporating additional data on hospice cost, this estimation procedure recovers the hospice cost function. The demand estimates show that reputation plays a significant role in consumer choice and depreciates at an annual rate of 46%. The hospice cost estimates show that an additional visit by a hospice costs around $192. Using the estimated demand and cost parameters, I solve for hospices' policy function via policy iteration. I use this to solve for equilibrium quality choices under various counterfactual policies.

My first set of counterfactuals involves changing the persistence of hospice reputation. A policy that increases the persistence of hospice reputation – for instance, a review website that centralizes information on quality provision – leads to higher hospice quality. Such a review website called "Hospice Compare" was introduced by CMS in 2017, suggesting that CMS has reached a similar conclusion. The website has seen insufficient activity from consumers; hence, my counterfactual suggests incentivizing consumer participation on this website can promote quality competition with little additional cost to CMS.

My second set of counterfactuals involves changing Medicare's reimbursement rates for hospices. I find that lower (higher) Medicare rates leads to firms choosing lower (higher) quality. Higher rates can also incentivize entry by new hospices, and the resulting competition further improves hospice quality; correspondingly, lower rates can induce exit and worsen quality further. Therefore, CMS should be cognizant of the effect of lower reimbursement on quality of service. Higher quality from higher reimbursement can also cause patients to switch away from even more expensive care, such as that documented in Gruber et al. (2023), potentially generating net savings for Medicare.

My third set of counterfactuals involves changing the reimbursement scheme structure for Medicare. I compare the current per-day Medicare reimbursement scheme with a hybrid per-day per-visit reimbursement scheme. The latter is found to achieve the same equilibrium quality level but with lower Medicare spending. Therefore, partially tying reimbursement to quality can lower spending without harming quality.

Contributions and related literature: First, I contribute to a very sparse literature on the hospice industry and its regulations. I am the first to structurally estimate the supply-side of this industry. Prior work on the hospice industry include Chung and Sorensen (2018), who estimate a nested logit demand model to examine market expansion from hospice entry; Dalton and Bradford (2019), who study length of stay of patients in non-profit and for-profit hospices; and Gruber et al.

(2023), who show that hospices save money for Medicare by offsetting alternative expensive care for Alzheimer's and Dementia patients.

Second, I am the first to build and estimate a structural dynamic oligopoly model of reputation accumulation through quality choice. I also construct a novel way of measuring reputation using market share data. This is in contrast with other papers that measure reputation with user reviews; my approach avoids the econometric issues involving user reviews and allows reputation to be studied in settings where such review data are not available. Very few papers attempt to analyze reputation acquisition via a structural model of firms. Apart from myself, two papers which do so are Saeedi (2019) and Bai (2024); my model and estimation are different from theirs. Saeedi (2019) studies how eBay sellers choose quantity over time to qualify for Powerseller status or Registered-Store status. In contrast to my paper, her model assumes quality of sellers is exogenous and buyers are unaware of past buyers' experiences. Bai (2024) studies watermelon sellers using an experiment where some watermelons are laser-branded as a costly signal of quality. Unlike my paper, she does not assume that sellers can precisely choose their quality. Her demand model explicitly accounts for consumer learning and beliefs while I do not. In contrast to this paper, she does not estimate the supply side of her setting. There is also an extensive theoretical literature on reputation; for an overview see Bar-Isaac and Tadelis (2008).

Third, I show the importance of reputation for patients choosing healthcare providers as well as healthcare providers choosing quality. This phenomenon is understudied in the health economics literature and potentially extends to other industries, particularly those with non-contractible quality or where a firm's brand capital plays a key role. There is a large empirical literature on reputation effects with most papers centered around online platforms, such as Cabral and Hortaçsu (2010), Chartock (2023), Fan et al. (2016), Newberry and Zhou (2019), and Tadelis (2016). There is much less work on reputation outside online platforms. Jin and Leslie (2009) study hygiene quality and reputation incentives in the context of restaurants; Bachmann et al. (2023) study spillover effects on reputation from the Volkswagen emissions scandal; and Sorensen (2006) studies social learning about health plan quality within university departments.

Fourth, I add to the literature on how healthcare providers choose quality. In contrast to other papers in this area, I show that quality choice has dynamic effects through influencing provider reputation. Papers on healthcare provider quality include Camarda (2022), Einav et al. (2018), Gaynor (2006), Gaynor et al. (2013), Grieco and Mcdevitt (2017), Hackmann (2019), Hotz and Xiao (2011), Lin (2015), Wang (2022), and Xiao (2010). Similar to my paper, Einav et al. (2018) study alternative reimbursement schemes that can generate financial savings without harming quality, in their case for long-term care hospitals.

A strand of literature that this paper relates to is the advertising and brand management literature in marketing. Several structural papers in that literature look at how firms spend money on advertising to build up their brand equity; see Dubé et al. (2005) and Borkovsky et al. (2017), and the references therein. This paper also adds to the literature on estimating dynamic games. I use the framework from Ericson and Pakes (1995), and use estimation methods from Bajari et al. (2007) and Pakes et al. (2007). Other papers on similar methods include Aguirregabiria and Mira (2007), Collard-Wexler (2013), and Ryan (2012); for detailed overviews see Doraszelski and Pakes (2007), Arcidiacono and Ellickson (2011), and Aguirregabiria et al. (2021). My paper has similarities with Benkard (2004), in that I use a nested-logit demand framework within a dynamic game of accumulation. While my paper is about reputation accumulation, Benkard (2004) is about accumulating experience levels. This paper also relates to the dynamic investment literature in industrial organization; classic references are Olley and Pakes (1996) and Ericson and Pakes (1995).

2 Industry background

Hospices provide palliative care to terminally ill patients, the majority of whom die within 1 week to 3 months in my dataset. Patients who enroll in hospice typically no longer receive curative treatment. Most patients in the dataset are covered by Medicare's reimbursement scheme, which pays hospices a fixed amount for every day the patient is enrolled. In return, hospice staff provide care to the patient by visiting her at her residence. I claim that a hospice can build up its reputation by consistently providing high quality over time. The absence of price competition or spatial competition makes hospice reputation particularly salient, and suggests that hospices compete mainly in reputation.

2.1 Hospice care provision

Unlike other medical providers, hospices typically provide care at the residence of the patient. Nurses and home health aides make the majority of hospice visits.[2] The hospice staff make multiple visits to a patient throughout the patient's enrollment period. Administrators of a hospice decide the number of visits, and set up the schedule for each nurse. A day for a hospice nurse involves driving to several patients as determined by the administrator, giving them care, writing up reports for the hospice physician and administrator, and refilling supplies.

The care given by hospice nurses is relatively low-skill compared to other medical providers. Hospices are not responsible for curative treatment; they are tasked with pain management and ease-of-living for a dying patient. Examples of hospice care include administering pain medication, providing medical supplies like oxygen and bandages, dressing bedsores, giving physical and speech therapy, helping with bathing and feeding, temporarily substituting as the primary caregiver, tending to emotional and spiritual needs of the patient, and giving grief counselling to the family after the patient passes away.[3]

It is important to note that the hospice decides how many visits to make to each patient and what care is to be provided during each visit. As is explained below, the patient does not incur any additional cost for more visits, and so would plausibly like as many visits as possible.

How a hospice is chosen by a patient and her family vary. Sometimes a social worker at the hospital gives the patient a list of hospices, and suggestions on which to choose based on

[2] Hospices employ several types of staff: registered nurses, licensed vocational nurses, home healthcare aides, physicians, social workers and chaplains. Physicians rarely make visits, and usually deal with reports on the patients from nurses. Social workers help the patient and her family with paperwork and choices.

[3] Hospice staff do not function as the primary caregiver of the patient - someone in the patient's residence (a family member if the patient resides at home, or a nurse if the patient resides in a nursing facility) has to be the primary caregiver. But hospice nurses assist the primary caregiver, and sometimes may substitute in for a period of time to give respite to the primary caregiver.

past experience; the same can be done by the patient's physician. The family can also search for hospices online (aided by online reviews and forums), or can rely on word-of-mouth from other families that have needed hospice services in the past.

2.2 Payers and reimbursement schemes

For hospice reimbursement, I focus my analysis on eligibility rules and reimbursement schemes for Medicare only. While hospice services can be paid for by a variety of payers – Medicare, Medi-Cal (California's Medicaid), private insurance, charity, self-pay, etc. – the overwhelming majority of patients are paid for by Medicare (see Table 1). Furthermore, Medi-Cal's reimbursement is essentially the same as Medicare's.

For a patient covered by Medicare, hospice care is essentially free. There is no out-of-pocket cost for enrolling into or receiving visits from a hospice. Patients have to give a copay of up to $5 for outpatient drugs required for pain management. This means that the "price" of choosing a hospice does not vary across hospices in the patient's choice set, and moreover can be thought of as zero. The patient generally has to turn down curative treatment to enroll into hospice care via Medicare.

For the hospice, enrolling a patient covered by Medicare results in the hospice getting paid a fixed rate for every day the patient is enrolled (i.e. a per-diem rate). It is important to note that the fixed rate is set exogenously and does not depend on the number of visits that the hospice staff makes. To contrast extreme examples, if a patient enrolls in a hospice for 10 days, the hospice gets paid a fixed amount for each of those 10 days irrespective of how many visits they make.[4][5]

The Medicare rate is determined as follows. Medicare sets a national per-diem rate that is split into a "labor component" and a "non labor component". Medicare also creates wage indices for each Metropolitan Statistical Area (MSA), such that a MSA with high wages will have a high wage index. The per-diem reimbursement rate of a hospice is the non-labor component and the labor component times the wage index of the hospice's MSA.[6] These rates are updated every year.

Payer	% of patients
Medicare	83.7
Medi-Cal	7.37
Private insurance	6.37
Selfpay	1.65
Charity	0.82

Table 1: Percentage of patients in dataset covered by each payer type.

2.3 Hospice competition, quality, and reputation

The institutional details indicate that hospices primarily compete in quality. Hospices do not compete in prices as they are paid a fixed rate by Medicare. Hospices also do not compete on location conditional on being in the same market – a patient's distance to the hospice is unlikely to play

[4]Medicare also distinguishes between different types of hospice care. Specifically, it draws a distinction between routine care (which is the care described above), inpatient care (where a patient moves to an inpatient ward of a hospice for intense treatment) and continuous respite care (where the hospice takes over as the primary caregiver). The majority of patients in my sample are given routine care; I show in Appendix C that of all the "days of care" provided by a hospice within each year, over 99% is dedicated to routine care. Medicare allows the per-diem rate to vary substantially based on whether the care provided is routine home care, inpatient care, or continuous care.

[5]There are additional dimensions of the Medicare reimbursement scheme that I do not study in this paper. First, the total revenue paid by Medicare to a hospice is subject to a cap that rises with the number of patients enrolled in the hospice. For a detailed study, see Gruber et al. (2023). Second, from 2017, Medicare began paying hospices an additional amount for visiting a patient in the last week of her life (called the "service intensity add-on"). Third, Medicare also lowers the per-diem rate for routine care if a patient is enrolled for more than 60 days to discourage hospices from enrolling non-terminal patients. Interestingly, it seems from news reports in 2019 that the service intensity add-on had seen limited uptake from hospices due to lack of understanding; see https://hospicenews.com/2019/05/20/medicare-service-intensity-add-on-underused-by-hospices/.

[6]To work out an example: for FY 2021, the labor component of routine home care is \$136.9, and the non-labor component is \$62.35. Suppose a hospice is in a county with a wage index of 0.8. The reimbursement rate of the hospice is approximately $136.9 \times 0.8 + 62.35 = \$171.87$. See Appendix J.1 for an example of Medicare's reimbursement rates and its variation across counties.

a role in her hospice choice as the staff visit the patient at her residence. Higher quality is more costly to provide, and hospices are not directly reimbursed for higher quality, so the main economic motive for providing higher quality is to attract more patients.

Furthermore, I hypothesize that hospices do not just compete in quality, but more specifically in reputation – which reflects both current and past quality choices. When looking for advice on how to choose a hospice, one common suggestion is to choose hospices that are reputable and have served the community well over a long time.[7] This is unsurprising as healthcare providers generally rely on good reputation to continue attracting patients. Thus we can expect reputation to matter for hospice choice.

This suggests that a hospice which has provided high quality of care in the past is more likely to be suggested by social workers, better known in the community, and better reviewed than a hospice which has not. Such a hospice is more likely to be selected by a patient and her family, and so will have greater market share. Presence of intermediaries such as social workers also suggests how information about a hospice's past quality choices can persist over time.

To confirm these ideas, I provide descriptive evidence of hospices competing in quality and reputation in the next section.

3 Data

The dataset for this paper comes from firm-level data on hospices in California for 2002-2018. A market is defined to be at the level of a county, and I restrict my estimation to 39 counties. Total firms in a market range from 1 to 23, with moderate entry and little exit. I use these data to

[7] Some websites which give suggestions on how hospices are chosen are listed in Appendix J alongside relevant quotes. A representative quote from AmericanHospice.org reads: "What do others say about this hospice? Get references both from people you know and from people in the field – e.g., local hospitals, nursing homes, clinicians. Ask anyone that you have connections to if they have had experience with the hospice and what their impressions are. Geriatric care managers can be a particularly good resource, as they often make referrals to hospices and hear from families about the care that was provided."

Similarly, HospiceFoundation.org states: "Seek professional opinions. Ask clinicians, professional caregivers at nursing homes, geriatric care managers, or end-of-life doulas about their experience with a hospice. Talk to friends, family, and neighbors who have used hospice services and get their opinions about the experience with a provider."

construct firm-level measures of quality, defined to be visits-per-patient-day made by a hospice in a year. I provide descriptive evidence showing that visits-per-patient-day is an appropriate measure of quality and suggesting the presence of reputation effects.

3.1 Data sources

The main dataset comes from Home Health Agencies And Hospice Annual Utilization Reports compiled by California Department of Health Care Access and Information (HCAI). Hospices in California are required to submit yearly utilization reports to HCAI where they report firm-level statistics for the year. This includes measures such as current location, total number of patients, total visits per year (broken down by type of staff), hospice characteristics, aggregate characteristics of patient pool, annual expenses, etc. I collect and clean utilization reports of hospices for 2002-2018. I complement this with additional data sources on population sizes (by age), mortality rates, and Medicare hospice reimbursement rates for California over the same time period.

Additional details on how I deal with nonresponses, outliers, and firm IDs are specified in Appendix A.

3.2 Market definition

A market is defined to be at the level of a county.[8] Market size is constructed by multiplying the population size of each age for a market with the corresponding mortality rate; that is, market size is the expected number of deaths happening in a market in a year.

I focus on 39 counties for my project. While California has 58 counties, I have to drop the remainder for the following reasons. First, several counties do not see any hospice presence over the course of my dataset. Second, a few counties have a very large number of hospices (Los

[8]While some hospices offer services to multiple counties, I can rule out clusters of counties as a market definition using data. Specifically, the dataset contains a breakdown of a hospice's patient pool by patients' counties of residence. Most of a hospice's patients originate from the county where the hospice is located; see Appendix C.4. The only exception are Yuba and Sutter counties which have a large overlap of patients, and so I combine them into a single market.

Angeles county has over 300)[9]. Markets with many small sellers have very different dynamics than those with few sellers. My model and estimation are oriented towards strategic interactions among small numbers of players and does not naturally accommodate such active markets. Furthermore, some of these markets exhibit such rapid growth in number of hospices that they are likely to be exhibiting nonstationary dynamics, which my stationary dynamic model is not well-equipped to handle. Such sample selection is in line with many dynamic oligopoly papers such as Collard-Wexler (2013) and Lin (2015). Please see Appendix C.4 for more details on market selection and a list of the counties I focus on.

3.3 Summary statistics on competition and patients

I now present some summary statistics on competition and patient composition within my sample. The tables below and in Appendix C show distributions of various key variables; the main take-aways are as follows. First, the vast majority of hospices have over 100 patients. This assuages worries that my key measure of quality - visits per patient day - is mostly driven by fluctuating patient severity, since we would expect that variations in patient severity would even out over a large number of patients over the course of the year. Second, the median firm count is 2 with the 75th percentile at 4, suggesting that an oligopoly setting is the correct mode of analysis instead of perfect competition. Third, while the total entry and exit over the 16 years of my data is significant, there is very little turnover in a market within an individual year. In my model, I will treat exit as effectively exogenous due to the lack of variation in my data. Fourth, the vast majority of patients have a length of stay between a week and 3 months; see Appendix C for more details. Finally, Appendix C also does a breakdown of patients by diagnosis, revealing that roughly 40% of the patients are suffering from cancer.

The dataset contains information on hospice-specific services and characteristics. To give a snapshot, in 2017 there were 171 hospices in my dataset; of these, 12 had an inpatient unit, 15

[9] These are also the counties that see high entry rates by new hospices, while my selected counties see moderate levels of entry. A separate paper by myself in progress (Alam (2025)) is studying hospice efficiency and entry-exit in all California counties.

Table 2: Distribution of market-level characteristics.

Variable	Min	10%	25%	50%	75%	90%	Max
Firm count	1.0	1.0	1.0	2.0	4.0	7.0	23.0
Entry count	0.0	0.0	0.0	0.0	0.0	1.0	5.0
Exit count	0.0	0.0	0.0	0.0	0.0	1.0	3.0
Medicare rates	134.48	151.88	178.29	212.41	255.25	284.52	375.92

Notes: Each observation is at the county-year level. Medicare per-day rates are inflation-adjusted.

offered special pediatric services, 2 offered adult day care services, and 95 were for-profit. The dataset also breaks down hospices by agency type: 110 hospices were free-standing, 27 also had a home-health unit, and 31 were hospital-based (a few hospices had agency type classified as "Other"). I control for these characteristics in my demand estimation.[10]

3.4 Measure of hospice quality

I discussed in Section 2.3 that hospices compete in quality, and more specifically reputation. I now explain the measure of quality that I will use in my estimation.

I claim that the frequency with which a hospice visits its patients (visits-per-patient-day) reflects the quality and effort the hospice provides. This is based on several arguments.

1. I provide descriptive evidence showing that visits-per-patient-day by a hospice both responds to the level of competition the hospice faces and correlates with the hospice having higher market share (Tables 4 and 5). This suggests visits-per-patient-day is a measure of quality, since hospice quality should rise when competition increases, and hospices choosing higher quality should get higher market share.

2. A hospice which makes more visits is constantly checking and adjusting to the patient's condition, can manage symptoms better, and is likely to arrive quickly in case of an emergency. Such a hospice is also able to provide more respite to the patient's primary caregiver, which

[10]The dataset also states how many patients of a hospice were discharged to a different hospice. The distribution of this variable as a share of total patients is reported in Table 23. Over 40% of hospices do not discharge any patient to a different hospice, and of the remainder the fraction is mostly less than 2%. In general, there is very little variation.

may be highly valued by the family.

3. Visits-per-patient-day is plausibly a good proxy for the general effort exerted by a hospice. Visits are the most costly part of care as the hospice has to hire extra nurses and pay the fuel cost of traveling. If a hospice is incurring this cost already, it is unlikely that they are then shirking on within-visit care. Therefore, it is also likely that number of visits is correlated with the general effort of the hospice, and so with other unobserved dimensions of quality.[11]

4. Visits-per-patient-day is in line with several measures of hospice quality tracked by CMS (see Appendix N for details). Among other indicators, CMS tracks: Hospice Visits in Last (three) Days of Life, Gaps in Skilled Nursing Visits, Skilled Nursing Care Minutes per Routine Home Care Day, Skilled Nursing Minutes on Weekends, Training Family To Care For Patient, and Getting Timely Help (which asks about hospice availability during evenings and holidays). These measures are all related to the frequency of visits made by a hospice.

   It is also reported that the above CMS indicators correlate well with survey-based measures of caregivers and patients. In fact, hospices have complained that CMS does not track non-nursing visits from the professional team, which my dataset actually reports. Furthermore, it is reported that "Gaps between nursing visits has been an area of concern for the OIG".[12]

   CMS also tracks "HIS Comprehensive Assessment at Admission", which measures the proportion of patients for whom the hospice performed the following 7 care processes: Beliefs/values addressed, Treatment preferences, Pain screening, Pain assessment, Dyspnea treatment, Dyspnea screening, and Treated with an opioid who are given a bowel regimen. We would expect that hospices which make more visits are also more likely to regularly perform these care processes, particularly those associated with screening and assessment.

   Note that while CMS collects quality data and family caregiver surveys, all the data collec-

[11] Hospice care mostly involves relatively low-skill care such as giving pain medication, making adjustments to living arrangements, and helping out the patient with certain physical activities. As such, it is unlikely that there will be much heterogeneity across hospices regarding what is done during visits.

[12] These comments are from the following report by NHPCO: https://www.cms.gov/files/document/hospice-care-index-hci-technical-reportjuly-2022.pdf

tion began on or after 2017 and is often very sparse. Thus, they cannot be incorporated into my main analysis of dynamic competition. I merged the CAHPS family caregiver ratings of hospices for the year 2018 with my data and ran some descriptive regressions. The results are very noisy and rely on a small sample size, so they should be interpreted with caution. I find that visits-per-patient-day is positively but weakly partially-correlated with ratings. I also find that visits-per-patient-day is more strongly related to market share than ratings. The results are discussed and reported in Appendix N.

Finally, there has been a change in the Medicare hospice reimbursement scheme since 2017 where hospices are reimbursed for visiting patients in the last week of their life, suggesting CMS wants to incentivize greater visits during that period.

5. I can compare my quality measure with those used for other providers in the literature. My measure aligns with the nursing home literature (Hackmann (2019), Lin (2015)), which measures nursing home quality by nurses-per-resident. Compared to the literature on other provider types, my measure has both advantages and disadvantages. The hospital literature uses readmission and mortality rates, while the dialysis literature uses infection rates (e.g. Grieco and Mcdevitt (2017)). Their advantage is that they measure outcomes; however, these are measures of extreme events – a hospital or dialysis center may provide poor quality yet not cause readmission, death, or infection in their patients. In contrast, my measure is input-based, and so it has the advantage that it picks up more subtle differences in quality and effort.

Therefore, my quality measure reflects how often a hospice visits its patients. The dataset gives me total visits that a hospice makes to its entire pool of patients within a year and the number of patients it serves in the same year. The data also contains a breakdown of length-of-stay for the patient pool in a hospice-year (how many patients stayed for 0-7 days, 8-30 days, 31-90 days, 91-179 days, and 180+ days). This allows me to construct an "average length of stay" for patients at a given hospice-year. Table 3 shows the distribution of these variables.

I combine the above variables to construct my measure of quality "visits-per-patient-day" as follows. Multiplying the total patients at a hospice with its average length-of-stay gives the total patient-days at the hospice. Dividing total visits[13] in a year by a hospice with its total patient-days gives the visits-per-patient-day.[14] To deal with high outlier values, I censor the visits-per-patient-day variable at the 90th percentile. That is, for values which are greater than the 90th percentile, I recode those values to be equal to the 90th percentile value.

Table 3: Distribution of visits and enrollment

Variable	10%	25%	50%	75%	90%
Visits per patient	21.0	26.67	33.68	44.84	61.43
Average length-of-stay	35.91	42.28	49.82	58.19	66.54
Visits per patient-day	0.48	0.57	0.69	0.84	1.06
Patients	63.0	149.0	289.5	511.75	765.0
Market share	2.77	7.37	19.71	40.04	100.0

Notes: For this table, "market share" denotes within-hospice market share, i.e. fraction of hospice patients in a market-year enrolling into a hospice. Each observation is at the hospice-year level.

I conduct multiple empirical checks to strengthen my claim that visits-per-patient-day is a good proxy for effort and quality. First, I regress visits-per-patient-day on the number of rivals and other covariates, and show that greater competition is associated with a hospice increasing its visits-per-patient-day. This aligns with the idea that hospices are choosing visits-per-patient-day to compete for patients (Table 4). Second, I regress market share on visits-per-patient-day and other covariates, and find that higher visits-per-patient-day is associated with higher market share (Table 5). This aligns with the idea that visits-per-patient-day is a measure of the quality that patients consider when choosing hospices.

I also study the persistence of quality choices over time, and how quality choices correlate with

[13] I aggregate all visits across staff type rather than select specific staff types, or allow different staff to contribute differently to the overall quality measure. This is because in my dataset, I see all staff types co-move very closely for hospices; see Section C.5 for Principal Components Analysis.

[14] It is often the case that patients need more visits near the end of their length of stay compared to earlier. Thus I construct the average length of stay to reflect this possibility. For the length-of-stay bins, I construct the chosen point in each bin to be 7, 25, 60, 100, and 230. Then the number of patients in each bin is multiplied with the corresponding point. The intuition is that patients in bins 0-7 days and 8-30 days need visits over most of their length of stay; however, longer staying patients may need visits for fewer days than their length of stay. I also estimate the demand system with visits-per-patient as the quality variable and obtain similar results.

Table 4: Regression of quality on level of competition.

	Quality			
	(1)	(2)	(3)	(4)
Count of rivals	0.037***	0.032***	0.029***	0.024*
	(0.005)	(0.008)	(0.007)	(0.009)
Count of rivals squared	-0.001***	-0.001**	-0.001**	-0.001*
	(0.000)	(0.000)	(0.000)	(0.000)
Reported average cost of visit			-0.140***	-0.141***
			(0.018)	(0.018)
County Fixed Effects	Yes	Yes	Yes	Yes
Year Fixed Effects				Yes
Controls		Yes	Yes	Yes
Obs.	2,066	2,066	2,066	2,066

Notes: Controls are hospice inpatient unit indicator, pediatric program indicator, bereavement for non-hospice-survivors indicator, day care for adult services indicator, for-profit status, and agency types indicators. Standard errors clustered at county-level in parentheses. Reported estimates and standard errors rounded to 3 decimal places.

average lengths-of-stay. First, I find that there is notable dispersion in the quality choices made by hospices going from one period to the next; this is illustrated in Figure 1. Looking ahead, this dispersion will inform the estimates in Table 5, and will help with the estimation of reputation effects in the demand model. Second, I find no partial correlation between visits-per-patient-day and average length-of-stay; this is illustrated in Table 16 and Figure 5. This assuages worries that visits-per-patient-day is driven by selection on lengths-of-stay.

3.5 Hospice reputation

Next, I show suggestive evidence that hospices accumulate reputation over time to gain market share. These will be based on two pieces of intuition. First, hospices which consistently chose high quality in the past have built up their reputation, and so should have higher current market share. Second, new entrants should see their market share grow over time as they build up their reputation. I find evidence for both in my dataset.

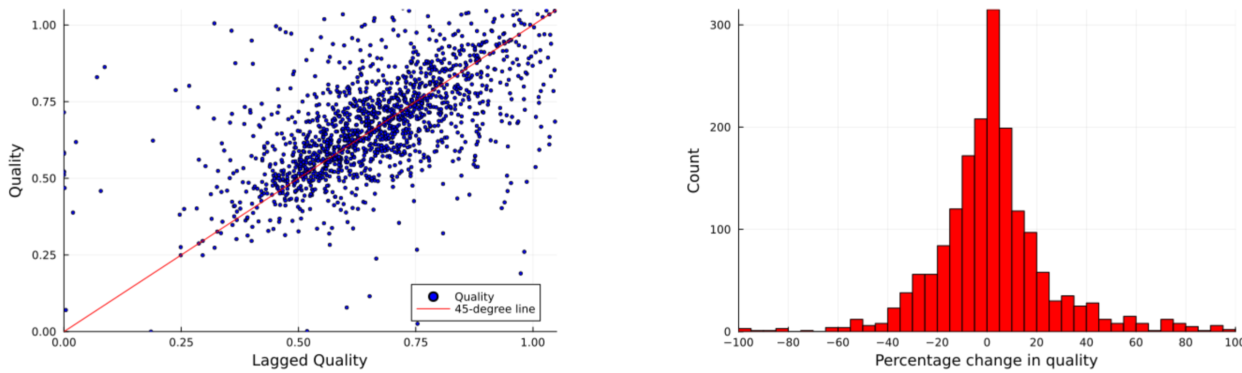
First, I show that past visits-per-patient-day impacts current market share. I regress a hospice's

Table 5: Regression of market share on quality and lagged quality.

	log(market share)				
	(1)	(2)	(3)	(4)	(5)
Count of rivals	-0.058***	-0.051***	-0.038***	-0.033*	-0.045**
	(0.011)	(0.010)	(0.009)	(0.013)	(0.013)
Visits-per-patient-day	1.138***	0.666***	0.558*	0.445*	0.450**
	(0.253)	(0.173)	(0.223)	(0.203)	(0.150)
Visits-per-patient-day$_{t-1}$		0.410**	0.366**	0.322*	
		(0.126)	(0.127)	(0.139)	
Visits-per-patient-day$_{t-2}$		0.449**	0.361**	0.236	
		(0.136)	(0.122)	(0.146)	
Visits-per-patient-day$_{t-3}$			0.496***	0.449**	
			(0.120)	(0.135)	
Visits-per-patient-day$_{t-4}$			0.272**	0.228**	
			(0.088)	(0.080)	
Visits-per-patient-day$_{t-5}$			0.084	-0.034	
			(0.160)	(0.171)	
Visits-per-patient-day$_{t-6}$			-0.041	-0.167	
			(0.151)	(0.158)	
Sum of past 3 quality choices					0.314**
					(0.100)
County Fixed Effects	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects				Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes
Obs.	2,066	1,597	937	937	1,403

Notes: The notation Visits-per-patient-day$_{t-n}$ denotes the visits-per-patient-day chosen n years prior. Sum of past 3 quality choices = $\sum_{i=1}^{3}$ Visits-per-patient-day$_{t-i}$. Controls are hospice inpatient unit indicator, pediatric program indicator, bereavement for non-hospice-survivors indicator, day care for adult services indicator, for-profit status, and agency types indicators. Standard errors clustered at county-level in parentheses.

Figure 1: Variation in quality choice across periods.



(a) Scatterplot of current versus lagged quality (censored observations excluded).

(b) Histogram of percentage change in quality from one period to the next.

current market share on its current and past visits-per-patient-day, as well as other covariates (Table 5). I find that lags of visits-per-patient-day have statistically significant and positive impacts on current market share, and improve the overall fit of the regression. In addition, I find that quality choices further in the past have generally smaller explanatory power, and after 4 years have no explanatory power anymore. This suggests that the effect of reputation declines over time, and thus informs my model setup.

Second, I show that new entrants' market shares grow in the years following entry, then level out over time. This is suggestive of reputation effects that decay over time: if reputation buildup helps a hospice attract patients, then new entrants should see growing market shares after entry as they accumulate reputation, and that growth should slow down with time. I limit my sample to hospices which enter within my dataset's time coverage, then regress their market share on firm fixed-effects, indicator of entry year, and indicators for years after entry. Table 6 shows the results. With the firm fixed-effects picking up the average market share of each hospice in my dataset, the years of entry indicator variables measure how market share adjusts to the average over time. I find that the market share of a new entrant increases over time, rather than adjust to the average immediately. While part of the deficit in the first year could be because the hospice may not have operated for the entire year, the pattern is present for subsequent years as well.

Table 6: Regression of market share on indicators for entry year and years after entry.

	log(market share)			
	(1)	(2)	(3)	(4)
Entry year	-0.916***	-0.942***	-0.952***	-0.960***
	(0.098)	(0.102)	(0.105)	(0.107)
Entry year + 1	-0.313***	-0.338***	-0.349***	-0.357***
	(0.067)	(0.070)	(0.072)	(0.078)
Entry year + 2		-0.110*	-0.120*	-0.128*
		(0.044)	(0.051)	(0.054)
Entry year + 3			-0.043	-0.051
			(0.040)	(0.044)
Entry year + 4				-0.040
				(0.046)
Firm Fixed Effects	Yes	Yes	Yes	Yes
Obs.	977	977	977	977

Notes: Standard errors clustered at county-level in parentheses. Entry year + n is an indicator variable for the year n after entry by a hospice.

3.6 Cost measures from hospice cost data

The data includes breakdown of annual hospice expenses. From this, for each hospice-year, I construct measures of per-period fixed cost, per-patient non-visit cost (hereafter referred to as "intercept cost"), and cost of making a visit (hereafter referred to as "visit cost"). All cost measures used in the analysis are inflation-adjusted. Details of variable construction, as well as a list of all the variables in the cost data, are listed in Appendix K.

4 Structural model

To study and model reputation in this setting, I set up a structural model of demand and supply for the hospice industry. The demand side is a discrete-choice model that incorporates a measure of reputation of the hospice. The supply side is a dynamic oligopoly model where forward-looking hospices are choosing quality in a competitive setting. The remainder of the section details the equilibrium concepts and the timing of the dynamic game.

4.1 Demand

The demand for hospices is modeled as a nested-logit discrete choice model, with a nest on hospices. A consumer can choose between any hospice present in the market in that period, or an outside option. The outside option of not choosing a hospice can be interpreted as the consumer continuing curative treatment, or choosing to be cared for by family/nursing home staff.

The utility of consumer i from choosing hospice j in period t is given by:

$$u_{ijt} = \alpha_{m(j)} + X'_{jt}\beta_d + \psi_{jt} + \xi_{jt} + \zeta_i + (1-\sigma_n)\tilde{\varepsilon}_{ijt} \tag{1}$$

$$\xi_{jt} = \rho\xi_{jt-1} + \varepsilon_{jt} \tag{2}$$

where $m(j)$ denotes the market in which hospice j is present and X_{jt} represents hospice characteristics[15]. The term ξ_{jt} denotes hospice-specific unobservables that are allowed to be serially correlated and vary over time. This can reflect factors like how connected the hospice is with physicians/nursing homes, advertising, past scandals, additional hospice services, etc. The term ψ_{jt} is the reputation stock of hospice j in time t. Finally, $\zeta_i + (1-\sigma_n)\tilde{\varepsilon}_{ijt}$ reflect that this is a nested logit – there are two nests, one on all hospices and one on the outside option.

Reputation is modeled as a stock transition equation:

$$\psi_{jt} = (1-\tau)\psi_{jt-1} + \eta a_{jt} \tag{3}$$

where a_{jt} is the visits-per-patient made by hospice j in period t. Assuming a new entrant has a reputation of zero, this can be rewritten as:

$$\psi_{jt} = \eta[a_{jt} + (1-\tau)a_{jt-1} + (1-\tau)^2 a_{jt-2} + \ldots]$$

[15]The hospice characteristics are dummy variables for the following: hospice inpatient unit, pediatric program, bereavement services, day care for adults, for-profit status, free-standing hospice, home-health-based hospice, hospital-based hospice, and hospices classified as having agency-type "Other".

That is, a firm's reputation is a discounted sum of its current and past choices of quality.[16]

4.2 Supply

To understand how hospices choose quality over time, I set up a discrete-time infinite-horizon dynamic oligopoly model. To do so, I combine the above demand system with cost function and payoff function to formulate a strategic dynamic optimization problem. This broadly follows the framework of Ericson and Pakes (1995). In subsequent sections, I lay out i) cost function, ii) per-period profit function, iii) dynamic programming problem, iv) equilibrium concept, and v) model-predicted choice probabilities.

In the dynamic game a period is defined to be a year. For the rest of this section the time subscript is suppressed for clarity unless necessary.

4.2.1 Cost function and per-period profit

The cost to a hospice for serving a patient depends on its quality choice and other cost components. I impose the cost to be linear in visits-per-patient-day.

The expected cost of serving each patient at quality a_j (henceforth referred to as marginal cost) is given by:

$$MC_j(a_j) = \gamma_{0j} + \gamma_{1j} a_j \overline{LOS} \tag{4}$$

where $\overline{LOS}$ is the expected length-of-stay for a patient. Thus $a_j\overline{LOS}$ captures the average number of visits made to a patient over her entire length of stay, γ_{1j} is the cost of an additional visit, and γ_{0j} captures the non-visit cost for each patient. The functional form captures the intuition that marginal cost is linearly increasing in quality a_j given $\gamma_{1j} > 0$. From before, I refer to γ_{1j} as the visit cost and γ_{0j} as the intercept cost.

[16] Note that the impact of a firms' age is already subsumed in my reputation transition equation (a firm who has lived longer has accumulated a larger number of past quality choices into its current reputation stock).

Combining the demand function with Medicare prices and market size, the per-period profit of hospice j in market m choosing quality a_j is:

$$\bar{\pi}(a_j, \mathbf{a}_{-j}, \mathbf{x}_m; \theta) = M_m s_j(a_j, \mathbf{a}_{-j}, \mathbf{x}_m)\left[P_m^{MCAR}\,\overline{LOS} - MC_j(a_j)\right] - \phi_j \tag{5}$$

where $\mathbf{x}_m$ denotes the state variables in market m, $\mathbf{a}_{-j}$ denotes the vector of realized actions of all rival firms in the market, M_m is the market size, and ϕ_j is the per-period fixed cost of hospice j. Medicare's per-day rate P^{MCAR} is multiplied with the expected length-of-stay $\overline{LOS}$, giving the expected revenue generated by a patient over their length-of-stay. This profit function is interpreted as maximizing the sum of expected profit per patient. This is because when choosing quality, a hospice does not know the exact realizations of length-of-stay for each patient it will enroll. In Appendix I.2 I derive this profit function and discuss alternatives.

For estimation, I impose additional restrictions on the cost parameters. I assume that the visit cost γ_{1j} is the same for all hospices ($\gamma_{1j} \equiv \gamma_1$), while γ_{0j} and ϕ_j varies by market ($\phi_j \equiv \phi_{m(j)}$ and $\gamma_{0j} \equiv \gamma_{0m(j)}$).

For the demand estimation I use the continuous measure of quality choice; however for estimating the supply side, I discretize quality choices into six tiers. This has the advantage of improving computational tractability (especially regarding counterfactuals) while still being a meaningful improvement of quality.

4.2.2 Dynamic choice of quality

The dynamic oligopoly game is modeled as a discrete-time simultaneous-move game over an infinite horizon. Every period (year) the incumbents decide what level of quality to provide. All consumers of a hospice receive the quality level chosen. Persistence of reputation means that current quality choices affect future sales, hence a forward-looking model is required; hospices also make this decision in a competitive setting, hence a model of strategic interaction is needed.

Hospice j's value function is:

$$V_j(\mathbf{x}_m, \varepsilon_j^a, \sigma; \theta) = \max_{a_j \in \mathscr{A}} \mathbb{E}\left[\bar{\pi}(a_j, \mathbf{a}_{-j}, \mathbf{x}_m; \theta) + \varepsilon_j^a(a_j) + \beta V_j(\mathbf{x}'_m, \varepsilon_j^{'a}, \sigma; \theta) \middle| a_j, \mathbf{x}_m, \sigma\right] \quad (6)$$

where θ is the set of structural supply-side parameters, $\beta \in (0,1)$ is the per-period discount rate, ε_j^a is a vector of choice-specific errors for hospice j, $\varepsilon_j^a(a_j)$ is the choice-specific errors for hospice j choosing action a_j, and σ is the equilibrium strategy profile of the hospice and its rivals (to be defined later). The symbol $\mathscr{A}$ denotes the set of possible quality choices (recall that we discretize quality into six tiers for the supply side estimation), and a_j is the quality level chosen from this set. The choice-specific error terms are meant to capture unobservables affecting a firm's quality choice. This choice-specific structural error term is known to the firm but unknown to the econometrician. In keeping with the literature and to help with tractability in estimation, I assume these shocks to be *i.i.d* and distributed Type-1 Extreme Value.

The above setup reflects two assumptions about the choice-specific shocks, following Rust (1987). First, the choice-specific shocks enter the payoff function additively. Second, the choice-specific shocks evolve independently of other state variables.[17]

4.2.3 Equilibrium and Conditional Choice Probabilities

This section describes the equilibrium concept used and how it leads to the conditional choice probabilities (CCPs) predicted by the model. These CCPs will later be matched to observed prob-

[17] Below I list these assumptions explicitly; the following mirrors the discussion in Lin (2015). Let the vector of common knowledge state variables in the current period be $\mathbf{x}$. The vector of common knowledge state variables and private information is denoted as $\mathbf{s} = (\mathbf{x}, \varepsilon^a)$. The state-to-state transition probability is $F(\mathbf{s}'|\mathbf{s}, \mathbf{a})$, where $\mathbf{a}$ denotes the vector of firm choices in the current period. Following Rust (1987), the two assumptions can be stated as:

1. Additive Separability: This imposes that the choice-specific structural error ε^a enters the per-period payoff function additively, i.e. $\bar{\pi}(a_j, \mathbf{a}_{-j}, \mathbf{x}_m, \varepsilon_j^a; \theta) = \bar{\pi}(a_j, \mathbf{a}_{-j}, \mathbf{x}_m; \theta) + \varepsilon_j^a(a_j)$.

2. Conditional Independence: This comprises two assumptions. One, $\mathbf{x}'$ is independent of ε^a after conditioning on $(\mathbf{x}, \mathbf{a})$. Two, ε^a is *i.i.d.* and evolves independently of other state variables. Mathematically, this can be written as:

$$F(\mathbf{s}'|\mathbf{s}, \mathbf{a}) = F(\mathbf{x}', \varepsilon^{a'}|\mathbf{x}, \varepsilon^a, \mathbf{a}) = F(\mathbf{x}'|\mathbf{x}, \mathbf{a}) F_\varepsilon(\varepsilon^{a'})$$

where $F_\varepsilon(\cdot)$ denotes the distribution of ε^a.

abilities to estimate the cost parameters. For this subsection, the notation for parameters θ is suppressed from the arguments for clarity.

In keeping with the literature on empirical dynamic games, I focus on pure-strategy symmetric anonymous Markov Perfect Equilibrium (MPE). Each firm's strategy thus depends only on current period's payoff-relevant state variables and its private choice-specific shocks; see Appendix G.2 for further explanation. Therefore a firm's strategy can be written as a mapping from current-period states and private choice-specific shocks to actions:

$$\sigma_j : \left(\mathbf{x}_m, \varepsilon_j^a\right) \rightarrow \mathscr{A}$$

Let the vector $\sigma = \{\sigma_j\}_{\forall j}$ denote the strategy profile of all firms in a market. I now show how this can be used to construct CCPs.

The choice-specific value function of action a in state $\mathbf{x}_m$, $\hat{v}_j(a, \mathbf{x}_m, \sigma)$, is the discounted sum of payoffs to hospice j from choosing action a before choice-specific shocks ε_j^a are observed. Let the ex-ante value functions $V_j(\mathbf{x}_m, \sigma)$ be Equation (6) with the choice-specific errors integrated out:

$$V_j\left(\mathbf{x}_m, \sigma\right) \equiv \int V_j\left(\mathbf{x}_m, \sigma, \varepsilon_j^a\right) dG\left(\varepsilon_j^a\right) \tag{7}$$

The ex-ante value function can be written in terms of choice-specific value functions:

$$\begin{aligned} V_j\left(\mathbf{x}_m, \sigma\right) &= \int V_j\left(\mathbf{x}_m, \sigma, \varepsilon_j^a\right) dG\left(\varepsilon_j^a\right) \\ &= \int \max_{a_j \in \mathscr{A}} \left\{\hat{v}_j\left(a_j, \mathbf{x}_m, \sigma\right) + \varepsilon_j^a\left(a_j\right)\right\} dG\left(\varepsilon_j^a\right) \end{aligned} \tag{8}$$

The MPE is the strategy profile σ^* such that every firm is choosing the optimal strategy given the strategies of their rivals:

$$\sigma_j^*\left(\mathbf{x}_m, \varepsilon_j^a, \sigma^*\right) = \arg\max_{a_j \in \mathscr{A}} \left\{\hat{v}_j\left(a_j, \mathbf{x_m}, \sigma^*\right) + \varepsilon_j^a\left(a_j\right)\right\}$$

Given this optimal strategy profile, the CCP of an action a_j taken by firm j can be written as:

$$\Psi_j\left(a_j, \mathbf{x}_m, \sigma^*\right) = \frac{\exp\left\{\hat{v}_j\left(a_j, \mathbf{x}_m, \sigma^*\right)/\theta_e\right\}}{\sum_{a \in \mathscr{A}} \exp\left\{\hat{v}_j\left(a, \mathbf{x}_m, \sigma^*\right)/\theta_e\right\}} \tag{9}$$

where θ_e is the logit scaling parameter of the Type-1 Extreme Value logit error ε^a. Since the choice-specific structural error is known to the firm but unknown to the researcher, I integrate out the errors and recover the CCP. The CCP therefore gives us the choice probabilities of all available actions for a given observed state.

Under the assumption of logit error, we can write the choice-specific value function $\hat{v}(a_j, \mathbf{x}_m, \sigma^*)$ as[18]:

$$\hat{v}_j(a_j, \mathbf{x}_m, \sigma^*) = \sum_{\mathbf{a}_{-j} \in \mathscr{A}_{-j}} \left\{ \left[\bar{\pi}(a_j, \mathbf{a}_{-j}, \mathbf{x}_m) + \beta V_j(\mathbf{x}'_m, \sigma^*) \right] F(\mathbf{x}'_m | \mathbf{x}_m, a_j, \mathbf{a}_{-j}) \prod_{n \neq j} \Psi_n(\mathbf{a}_{-j}[n], \mathbf{x}_m, \sigma^*) \right\} \tag{10}$$

where $\Psi_n(\cdot, \mathbf{x}_m, \sigma^*)$ is the CCP of firm n, $\mathscr{A}_{-j}$ is the set of all possible rival action profile vectors, and $\mathbf{a}_{-j}[n]$ is the action by firm n.

The logit assumption also allows us to rewrite Equation (8) as:

$$V_j(\mathbf{x}_m, \sigma^*) \quad = \quad \theta_e \left[0.577216 + \ln\Big(\sum_{a_j \in \mathscr{A}} e^{\hat{v}_j(a_j, \mathbf{x}_m, \sigma^*)/\theta_e} \Big) \right] \tag{11}$$

4.3 Timing

In any period, the industry state comprises common-knowledge variables (own characteristics, rival characteristics, market characteristics) and private information (choice-specific structural errors ε^a). The industry state comprises both endogenous and exogenous variables, and as a result evolves due to a mix of firm decisions and exogenous transitions.

[18]Prior to knowing the choice-specific structural errors, $\hat{v}(a_j, \mathbf{x}, \sigma^*)$ is hospice j's expected discounted sum of payoffs from choosing action a_j when facing state variables $\mathbf{x}$. Forming this expectation requires integrating over several random variables. The last term shows that this expression requires integrating over rival actions, which is done using the CCP at equilibrium strategies. Intuitively, since this is a simultaneous-move game, when hospice j chooses an action this period it does not know its rivals actions, and so has to make predictions using the equilibrium strategies. The middle term shows that conditional on own choice and rival choices, the firm also needs to integrate over all possible values of state variables. The first term is then the current period payoff and continuation value conditional on state variables and rival choices.

Given the model above, the game's evolution and timing is as follows. For period t and market m:

1. Incumbents observe $\mathbf{x}_{mt}$ and structural errors, and each make quality choices.
2. Reputation stock of each incumbent evolves.
3. Consumers observe $\mathbf{x}_{mt}$, reputation stocks, and structural errors, then choose a hospice.
4. Market shares are realized, and incumbents receive profits.
5. Incumbents stay or exit the market.
6. Potential entrants enter or disappear based on realized $\mathbf{x}_{mt}$.
7. All state variables evolve.

5 Discussion

Below I offer a more thorough discussion on reputation and the unobserved demand shocks in the discrete-choice demand model. See Appendix D for a discussion on capacity constraints and why it is unlikely to play a role in my setting; Appendix L for a heuristic discussion on identification of demand and cost parameters; and Appendix M for additional discussions on advertising and reputation index.

5.1 Reputation

I now discuss how reputation operates in my setting and explain the motivations behind the reputation transition equation (Equation (3)).[19]

[19]Note that instead of modeling reputation formation as a complex process of belief updating, I approximate the reputation stock of a hospice as a parametric function of its current and past quality choices. This reputation stock enters a consumer's utility function and influences her choice of a hospice. This parametric approximation allows me to study reputation using only firm-level data and maintain computational tractability. It also allows me to be agnostic about the specific channel through which reputation operates. Consumers may care about reputation because of the corresponding goodwill and name recognition, or because reputation is a good statistic for predicting future quality in the face of bounded rationality.

In my model consumers care about a hospice's reputation. Holding everything constant, a consumer is more likely to choose a hospice with high reputation over one with low.[20] Given this setting, why do consumers care about a hospice's reputation? A few reasons are:

1. A hospice with higher reputation has more goodwill and recognition in the community, and so is more likely to be referred to the patient. This has parallels with the intangible brand capital literature - for example Bronnenberg et al. (2022) talks at length about brand equity and reputation, and how these create an incentive for firms to repeatedly supply high quality over time. It also has parallels with reputation in a repeated game of quality choice under complete information.

2. To predict quality choice, consumers can track a firm's reputation alone instead of all possible state variables. As can be seen in the model above, quality choice is affected by factors such as costs, unobserved demand shocks, and hospice characteristics, all of which are serially correlated. Instead of keeping track of all these variables and their evolution, consumers can use reputation as a statistic for predicting the quality they will receive from the hospice. This reputation index can be learned by talking to social workers, physicians, and other information sources.

Another element of the model is that reputation depreciates at the rate of τ. Why does reputation depreciate? There are several possibilities. One, information on past quality choices disappears over time. This might be because of forgetting by the community. It could also be due to retirement or relocation of physicians, social workers, and community members. The patient's family is less likely to require a hospice soon after the patient passes away, so the fact that there are no repeat consumers means information is less likely to persist. Another way of thinking about this is that information on past choices is still available but gets more costly to retrieve over time.

[20] Using the terminology from the theoretical literature on reputation, my model is somewhat similar to a hidden action model of information (as opposed to hidden information). Hidden action models assume that quality is chosen by the firm and consumers try to predict a firm's quality choice with full information about its past choices. In contrast, a hidden information model assumes that a firm's quality is exogenously fixed, and the firm can choose to signal this quality type to consumers.

Two, past quality choices may be less relevant over time. This is because past quality choice also represents old and outdated medical practice styles, technology, and medicine. An example is that a hospice nurse's visit 20 years ago had very different content compared to a hospice nurse's visit at present. Third, quality choices very far in the past reflect the hospice's demand shocks, costs, and characteristics that may no longer have any predictive power, since these are also changing over time.

Finally, note that my reputation transition equation includes current quality as well as past quality. This means that current period reputation – which influences hospice choice by a consumer – is affected by current quality choice. This seems to indicate that consumers observe current quality choice; if so, why would they need to predict current quality choice using past choices? First, the decision to include current quality into reputation is mainly driven by the fact that my data is at the yearly level. This means information on current quality choice has time to diffuse through the community over the course of a year. Second, it subsumes models of quality competition in most empirical papers that include only current period quality in the demand system (e.g. Hackmann (2019), Lin (2015)). Finally, it accounts for the possibility that a hospice might be able (to some extent) to convince a potential consumer about the level of quality they will provide.

5.2 Unobserved demand shocks

The AR(1) process governing ξ_{jt} captures persistent unobservables in demand. One example could be that a hospice is well-connected with physicians and social workers in some hospitals and so get repeated referrals despite not choosing higher quality - this will manifest as the hospice having high ξ_{jt} for several periods. Another example could be a hospice that has suffered a recent scandal such as elder abuse. Despite choosing high visits-per-patient-day the negative publicity may last for several periods - this will manifest itself as the hospice having low ξ_{jt} for several periods.

While the above can be achieved without explicitly modeling ξ_{jt} as persistent, the AR(1) structure presents an additional advantage. It allows for the possibility that a hospice with high ξ might face less entry because it is more difficult to compete against. Without it, I would have to assume that level of competition – which is my main source of instruments – in the market is orthogonal to the ξ of all incumbents. But that might not be true: if the incumbent has persistently high (or low) ξ, then that could affect entry/exit decisions and hence the level of competition in the market. By allowing for persistence, the claim is that level of competition is orthogonal to some residual term after accounting for persistence. In the demand estimation this means I build moments around ε_{jt} instead of ξ_{jt}; this follows Grennan (2013) and Hodgson (2023).

I now discuss the distinction I make between reputation (ψ_{jt}) and unobserved demand shock (ξ_{jt}) of a hospice. Reputation is built up through a hospice's efforts over time, and my measure of visits-per-patient-day attempts to proxy for this effort. The unobserved demand shock is not interpreted as capturing unobserved quality or effort dimensions, but rather captures non-quality dimensions such as connectedness with physicians, scandals, factors outside the hospice's control, etc. As a result, ξ_{jt} is not interpreted as contributing to a hospice's reputation.

That said, to the extent that there are dimensions of quality not captured by my quality measure that help build reputation, those will likely get subsumed in the AR(1) process of ξ_{jt}. While this is hopefully less of an issue in hospice care, where the type of care provided is mostly low skill, it is nevertheless something I cannot conclusively rule out. If better data were available on the various dimensions of quality, I could include those to create a more comprehensive quality measure for my structural model.

Notably, when I regress estimated ξ_{jt} on visits-per-patient-day, I find no correlation (Table 14). This makes it less likely that ξ_{jt} is an unobserved quality dimension strategically chosen by the hospice. If a hospice were choosing visits-per-patient-day and ξ_{jt} as two separate dimensions of quality, it is unlikely that they would be selecting each in such an independent manner.

Finally, I reestimate demand without imposing the AR(1) structure on ξ_{jt}, thus building moments around ξ_{jt} instead of ε_{jt}, and continue finding evidence of reputation (Table 13).

6 Estimation and Results

This section details how the structural model above is estimated and the estimation results. The demand model is estimated using two-stage GMM following the inversion technique of Berry (1994). The supply side is estimated by leveraging additional cost data and using the method of Bajari et al. (2007) that matches simulated choices with observed choices. Demand estimates show that consumers are more likely to choose a hospice with higher reputation, and the reputation stock depreciates by 46% every year. Supply side estimates show that an additional visit by a hospice to a patient costs around \$192. Additional details about the estimation procedures are given in Appendix G.

6.1 Demand estimation

Using Berry (1994), the nested-logit discrete choice model can be rewritten in terms of observed market shares:

$$ln(s_{jt}) - ln(s_{0t}) = \alpha_{m(j)} + X'_{jt}\beta_d + \sigma_n ln(s_{j|gt}) + \xi_{jt} +$$
$$\eta[\mathrm{a}_{jt} + (1-\tau)\mathrm{a}_{jt-1} + (1-\tau)^2\mathrm{a}_{jt-2} + \ldots] \quad (12)$$

where s_{jt} and s_{0t} represent the market shares of hospice j and the outside option respectively, and $s_{j|gt}$ is j's within-hospice market share. Recall that for flexibility we also allow the unobserved demand shocks to be persistent:

$$\xi_{jt} = \rho\xi_{jt-1} + \varepsilon_{jt}$$

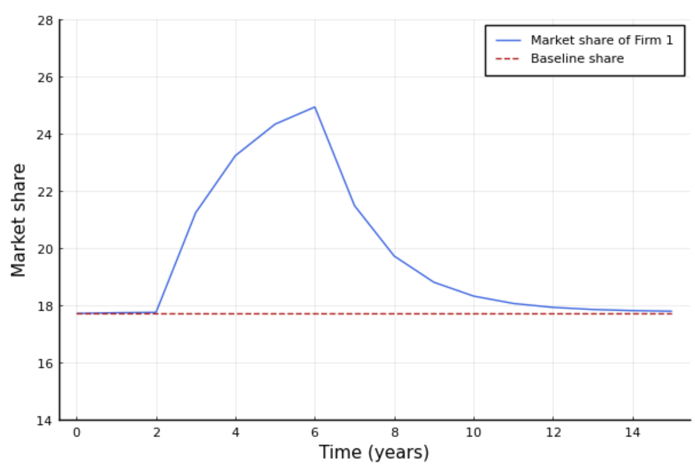
These equations are estimated simultaneously via two-step GMM. Instruments are multiplied with the error term ε_{jt} to give moment conditions for GMM. Further details of demand estimation are given in Appendix G.1.

Next I discuss endogeneity concerns and choice of instruments. Within-hospice market share

$s_{j|gt}$ is mechanically correlated with unobserved demand shock ξ, and it may be that reputation is also correlated with ξ. As a result, I use instruments from Berry et al. (1995) (henceforth BLP IVs) and county-dummy-by-fuel-prices (since I have fuel prices at the California level). BLP IVs are standard instruments in the IO literature that use measures of competitiveness in product space to create variation in within-hospice market share and quality choice while being orthogonal to ε_{jt} in the ξ transition equation. In my setting, these are the sum of rivals and sums of rival characteristics. Fuel prices are cost shifters that affect quality choice while being orthogonal to ξ.

Table 7 shows the results from the demand estimation. The reputation decay rate τ is estimated to be 0.46, which can be interpreted as saying 54% of a hospice's reputation transfers over to the next year. A positive value of η means that a hospice with higher current and past quality choices is more likely to be chosen by consumers; in other words, a hospice with higher reputation will have higher market share. The unobserved demand shocks are highly persistent (ρ), and consumer preferences within a nest are highly correlated (σ_n). The remaining values in Table 7 gives consumer preferences for hospice characteristics.

Figure 2: Impulse response in a duopoly setting to illustrate reputation accumulation.



Notes: Firm 1 and 2 choose identical quality ($a = 0.68$) in every period except periods 3-6, when Firm 1 chooses higher quality ($a = 0.92$). "Baseline share" denotes what Firm 1's market share would have been had it kept choosing Quality = 0.68 over the same period. Note that there is an outside option so the market shares of the two firms do not sum to 100.

To more easily interpret the value of η and τ, Figure 2 shows an impulse response to convey the importance of reputation effects. The setting is a duopoly. Firm 2 chooses 0.68 every period.

Table 7: Results of demand estimation.

	Demand
τ	0.463
	(0.164)
ρ	0.781
	(0.076)
σ_n	0.554
	(0.063)
η	0.540
	(0.155)
Hospice inpatient unit	0.044
	(0.133)
Pediatric program	0.253
	(0.078)
Bereavement services	0.012
	(0.041)
Day care for adults	0.090
	(0.178)
For-profit	-0.232
	(0.117)
Agency type: free-standing	-0.204
	(0.150)
Agency type: home health based	-0.346
	(0.172)
Agency type: hospital-based	-0.290
	(0.150)

Notes: The demand model is estimated using 2-step nonlinear GMM with BLP IVs and county-dummy-by-fuel-prices. Robust standard errors are in parentheses. Estimates of county fixed effects are suppressed. See Appendix G.1 for more details.
Also note that the absence of reputation persistence would mean $\tau = 1$; therefore the relevant t-stat for τ is $\frac{\hat{\tau}-1}{s.e.(\tau)}$.

Firm 1 chooses 0.68 every period except periods 3-6, when it chooses 0.92. That is, firms 1 and 2 make identical quality choices except in periods 3-6, when firm 1 chooses higher quality.

Figure 2 traces out the market share of firm 1. As firm 1 increases its quality in period 3, its market share rises, but reputation effects mean that its market share keeps rising in the next few periods even as it chooses the same quality level. This shows how reputation accumulates over time. After period 6 it returns to the original quality choice, but its market share remains elevated for several more years and gradually decays down to the original level. This illustrates how reputation decays over time if not maintained with high quality choices.

I conduct additional robustness checks for demand estimation, involving Gandhi-Houde IVs, reported average cost of a visit, subsets of instruments, not imposing the AR(1) structure on ξ_{jt}, etc. All specifications yield the same qualitative results; see Appendix B.

6.2 Cost function from cost data

From the cost data I construct hospice-by-year estimates of per-period fixed cost, intercept cost, and visit cost (adjusted for inflation rates). I regress each on county dummies to obtain county-level averages. Table 8 shows the distribution of regression coefficients, while Table 32 shows the full regression results.

Table 8: Distribution of coefficients from regression of cost on county dummies.

	10%	**25%**	**50%**	**75%**	**90%**
Fixed Cost	213,449.68	360,520.09	534,721.52	919,459.91	1,142,199.24
Intercept	257.40	372.09	587.18	794.80	993.89
Slope	85.82	92.37	109.65	131.13	143.21

Notes: Detailed results are presented in Table 32.

6.3 Estimation of dynamic oligopoly model

As an alternative to using the visit cost estimate (γ_{1j}) from the cost data, I recover both the visit cost estimate and the logit scaling parameter for the hospice's quality choice using the two-stage estimator from Bajari et al. (2007).[21] The first stage involves getting reduced-form estimates of the firms' policy functions from the data, as well as state transition probabilities. In the second stage, these policy functions are used to conduct forward simulation and generate model-predicted

[21] There are two points to note about this. First, the reason I use a two-stage estimator instead of a full-solution method is because of the presence of a large number of state variables (see below). This renders a full-solution method computationally infeasible. Second, a two-stage estimator assumes that a single equilibrium is played across all markets, and so cannot allow for multiple equilibria. Fortunately this is not an issue in my setting, because my model setup and institutional details rule out multiple equilibria; see Appendix G.5.

conditional choice probabilities (CCPs, explained in Section 4.2.3) for a given guess of cost parameters. Moment conditions are then constructed, and an optimizer searches to find cost parameter values that minimize the distance between model-predicted CCPs and observed probabilities. See Appendix G.2 for more details about implementation.

To simplify computation and model solution, I discretize quality choice into 6 tiers. Visits-per-patient-day is discretized into 6 bins between 0.44 to 1.04 visits-per-patient-day (open lower intervals and closed upper intervals) with increments of 0.12. From Table 3 it can be seen that this approximately covers the 10th to 90th percentiles of the observed visits-per-patient-day distribution.

I do not estimate the per-period fixed cost and intercept cost in the dynamic game estimation, instead fixing them at the estimated values from the cost data (see Tables 8 and 32). These are allowed to vary by county; following the notation from earlier, $\phi_j \equiv \phi_{m(j)}$ and $\gamma_{0j} \equiv \gamma_{0m(j)}$, where $m(j)$ is the county where hospice j is located. Per-period fixed cost is not identified since I do not model exit, and in simulations I found that the cost function intercept was not clearly estimated either.

The first stage of Bajari et al. (2007) requires an estimate of the policy function from the data. Broadly speaking, I regress quality on a variety of covariates to see how quality choice is affected by state variables. This sheds light on how hospices react to reputation levels and product characteristics of themselves and rivals. I interpret this regression as reflecting the policy function used by hospices when making quality decisions.

More specifically, I obtain an estimate of the policy function in two different ways. For the baseline case, I use a 2SLS regression (Table 26) to account for potential serially correlated unobservables following the ideas of Berry and Compiani (2021). As an alternative, I use a flexible regression (using third-degree polynomials of state variables) to approximate the policy function (Table 29). See Appendix E for details on how the empirical policy function is estimated, the results of the estimation, and robustness checks on the 2SLS regression by including cost data as covariates.

Note that while I use this empirical policy function for the estimation, I do not use it for my counterfactuals; instead, I will numerically solve for the optimal policy function in a simplified setting.

The state variables that enter a hospice's value function are its hospice characteristics and unobserved shocks, its rivals' characteristics and unobserved shocks, its lagged reputation, its rivals' lagged reputations, Medicare reimbursement rates, fixed costs, intercept costs, market size, and identity of the county where it operates.

Next I explain how I implement state transitions in my forward simulation. For the estimation, I assume for simplicity that a firm's characteristics do not change over time; this assumption can be relaxed. Firms' current period unobserved shocks (ξ) are imputed from the demand estimation; the estimated persistence rate is used to simulate it forward while drawing the errors (ε) from the econometric error distribution. Market size and Medicare rates follow an AR(1) process that is estimated from the data. From my data I find the average length-of-stay $\overline{LOS}$ to be 49.16 days, and I set the discount rate $\beta = 0.95$.

Entry is moderate and exit infrequent in my data, so I implement entry and exit as exogenous transitions rather than endogenizing them as part of the firm's choice problem. In my forward simulation, entry by potential entrants into a market is modeled through an ordered logit specified in Table 30. Exit by a firm happens with a probability of 4% every period, which matches exit rates in my data.

The first-stage policy functions and transition matrices are used in forward simulation to construct choice-specific value functions. This combines methods discussed in Bajari et al. (2007) and Pakes et al. (2007). See Appendix G.2 for a description of the estimation procedure and Appendix A for an explanation of how I impute reputation stocks of older hospices.

The simulated path continues for 80 periods. For each action and each observed state I run the forward simulation 200 times. The resulting choice-specific value functions are then used to construct the model-predicted choice probabilities.

Next, I describe how these model-predicted choice probabilities are used to estimate the hospice cost function. The intuition is that I search for values of parameters θ that minimize the distance between model-predicted choice probabilities and observed choices.

The dynamic oligopoly model is estimated using two-step GMM. The GMM residual term for observation n is:

$$\Xi_n(\theta) = a_n^{data} - \sum_{a_n \in \mathscr{A}} a_n \hat{\Psi}(a_n, \mathbf{x}_n^{data}, \sigma^*; \theta)$$

where $\hat{\Psi}(a_n, \mathbf{x}_n^{data}, \sigma^*; \theta)$ is the predicted choice probability for action a_n, a_n^{data} is the observed quality choice for observation n, θ is the guess of structural parameters, and $\mathbf{x}_n^{data}$ is the market state for observation n. The error term measures how close the average model-predicted choice is to the observed choice. The instruments Z used in GMM are the county indicators and the variables in the first-stage empirical policy functions (with the exception of the firm fixed effects), giving us the moment conditions $E[Z'\Xi_n(\theta)]$. The intuition for this is that the model's predicted choices leverage these variables, and so at the true parameter values the model's prediction errors should be orthogonal to these variables. The resulting GMM optimization problem is:

$$\min_{\theta} \left[\frac{1}{N}\sum_n Z_n' \Xi_n(\theta)\right]' \hat{W} \left[\frac{1}{N}\sum_n Z_n' \Xi_n(\theta)\right]$$

This objective function is minimized to estimate cost parameters of hospices (γ_1 and θ_e). Standard errors are calculated via block-bootstrapping, similar to Wang (2022).

Table 9 gives the results of the dynamic supply estimation.[22] I find that the cost of an additional visit is approximately \$192.

I now give a brief discussion of the estimate of γ_1. First, this estimate is higher than my measure of visit cost from the cost data (see Table 8), which had a median of \$109.65. The higher value from Bajari et al. (2007) estimation is reasonable because it measure the economic cost, while the cost measures from expense data reflect accounting cost. Economic cost includes, in addition to accounting cost, factors such as opportunity cost and presence of constraints. In my policy

[22]While Table 9 shows that θ_e is noisily estimated, I find that the objective function is convex in γ_1 and θ_e.

counterfactuals I will therefore use the cost measure from the Bajari et al. (2007) estimation.[23] Second, I interpret the cost estimate from the dynamic game estimation as averaging the economic cost of a visit over counties and hospices, as the accounting estimates suggest some heterogeneity. I could alternatively run my counterfactuals using the accounting cost estimates, and preliminary runs suggest very similar qualitative results. Note that this paper's aim is to estimate a reasonable approximation of the cost function to model hospice competition; a separate paper by myself (Alam (2025)) is focusing on hospice cost structure and efficiency differences using accounting cost reports.

Table 9: Results of supply estimation.

	2SLS	Flexible
γ_1	192.252	204.445
	(42.332)	(25.622)
θ_e	1.066×10^6	2.615×10^6
	(5.410×10^6)	(3.707×10^6)

Notes: The visit cost γ_1 and logit scaling parameter θ_e are estimated using the method of Bajari et al. (2007). Standard errors (in parentheses) are calculated via bootstrapping at the county-year level. 400 bootstrap runs were used. "2SLS" refers to the estimation using a 2SLS-approximation of empirical policy function, while "Flexible" refers to the estimation using a third-degree polynomial approximation of the empirical policy function.

7 Policy counterfactuals

Given the above estimates of the structural parameters, I can now conduct counterfactuals to investigate the effects of various policy tools. As reputation becomes more persistent, hospices choose higher quality. Lower Medicare reimbursement leads to worsening hospice quality. Compared to Medicare's current per-day reimbursement scheme, a hybrid per-day per-visit hospice reimburse-

[23] The estimate of the cost parameter may also subsume difficulties that hospices may face in hiring nurses given reports of nursing shortages in the US. That said, evidence suggests that the nursing shortage in California worsened considerably after 2020, which my dataset does not cover.

I also interpret my cost estimates as subsuming any altruism by the hospice. That is, greater altruism by a hospice will lead to lower cost estimates – the hospices will be choosing higher quality than would be suggested by their accounting costs, and my estimation will rationalize this as the hospice having lower cost. See Hackmann (2019) for a paper that backs out the altruism parameter from the preference function.

ment scheme achieves the same equilibrium quality with lower Medicare spending.

7.1 Solving for equilibrium outcomes

To conduct counterfactuals I need to solve the model and recover the policy function of the hospice. This requires me to calculate $\Psi(a_j, \mathbf{x}, \sigma^*)$, $\hat{v}(a_j, \mathbf{x}, \sigma^*)$ and $V(\mathbf{x}, \sigma^*)$ such that Equations (9), (10), and (11) simultaneously hold at every point in the state space. I solve the model using the policy iteration method (Rust (1994), Judd (1998), Benkard (2004), Caoui and Steck (2023)). This gives me the policy function and ex-ante value function at every action-state combination; thus, I solve for the optimal policy function and value function rather than using the empirical estimates from the data. For more details on model solution, see Appendix G.4.

It is not computationally feasible to solve the model with the full set of state variables and multiple dynamic decisions. Thus, to ease computational burden, I solve a simpler version of my setting. First, I solve the model for a representative market instead of for each market in my dataset. Second, I limit the number of firms to 2 or 3. This is a good approximation of the competitive environment in my data, since firm-count at the market-year level has a median of 2. Third, I do not allow for entry. This rules out firms acquiring reputation for entry deterrence. This is a reasonable assumption to make because I do not find strong evidence of entry deterrence through reputation in my data (I regress entry numbers on reputations of incumbents and do not find a relationship), as well as because entry happens in less than 10% of market-year observations. Finally, I do not allow for exit, which again is infrequently observed in the data.

While not explicitly modeling entry or exit, I report the ex-ante value functions of incumbents for all policy simulations. Changes in ex-ante value functions can be indicative of entry-exit incentives; for instance, a policy change that lowers ex-ante value functions may induce exit, while an increase may incentivize entry. In Appendix I.1 I discuss the intuition behind this approach as well as what it rules out.

Another decision to simplify computation is to discretize the reputation stock, and impose a stochastic reputation evolution process to minimize the effects of any such discretization. This closely follows Benkard (2004). I discretize reputation into bins from 0.0 to 1.5 with increments of 0.3; this range matches the reputation stocks observed in my dataset, and finer discretizations have yielded the same results.

The stochastic reputation evolution works as follows. Fix a firm, an industry state, and a chosen action a_t. Let ψ_t be the reputation stock predicted by the reputation transition equation $\psi_t = (1-\tau)\psi_{t-1} + \eta a_t$. Let ψ_d be the largest reputation stock less than ψ_t and ψ_u be the lowest reputation stock greater than ψ_t. In the counterfactual, the attained reputation stock ψ_t^* then follows the stochastic process below:

$$\psi_t^* = \begin{cases} \psi_u & \text{w.p. } \frac{\psi_t - \psi_d}{\psi_u - \psi_d} \\ \psi_d & \text{w.p. } 1 - \frac{\psi_t - \psi_d}{\psi_u - \psi_d} \end{cases} \tag{13}$$

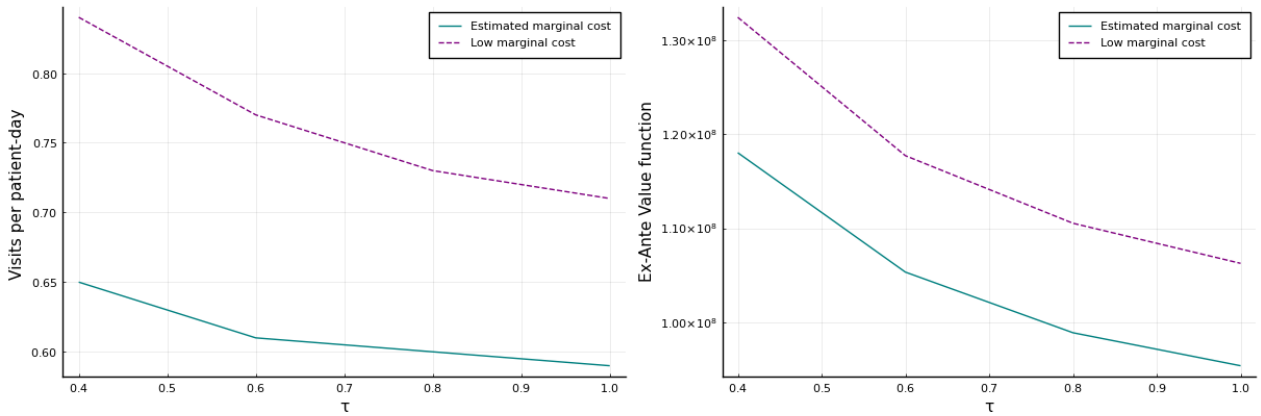
Counterfactual results are reported as follows. For a given policy experiment, I solve the model to obtain the policy function. The policy function gives the probability of a firm making each action choice for any given state under the chosen policy experiment. I pick a starting point for a market, then simulate forward 1000 periods using the policy function. I report the average quality choice by one firm over the simulation. To prevent my choice of initial conditions from contaminating the result, the first 200 periods are dropped; additionally I check by visual inspection to ensure that the industry has settled into its long-run state.[24] For visual illustration of the policy functions, see Appendix H.

7.2 Increasing persistence of reputation

The first set of counterfactuals involve the persistence of reputation. From the perspective of policymaking, this can tell us whether policy changes that lead to greater reputation persistence can improve consumer welfare. For instance, in 2017 CMS launched Hospice Compare, a website

[24] While I show how different policy levers can affect equilibrium quality, I do not specify what the quality target should be. This is because my demand model is silent about consumer valuation of visits. The thought experiment is that CMS has figured out the quality target through studies of their own, and can achieve said target with the policy instruments I explore.

Figure 3: Quality choice in a duopoly against τ for different marginal costs.



(a) (b)

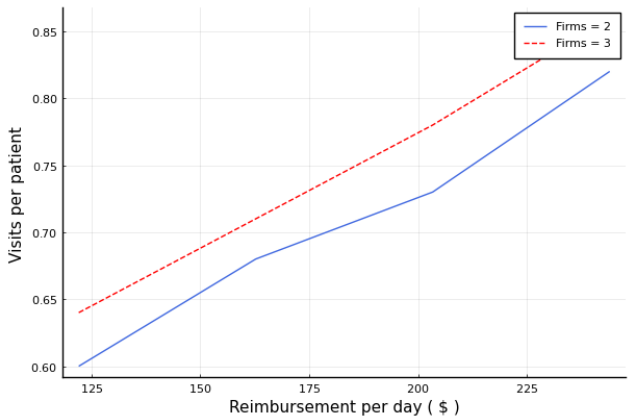
Notes: The figure reports the average quality choice and average ex-ante value function of one hospice over the simulation. The model is solved for two identical hospices via policy iteration.

that details quality information about hospices. Such review websites can aggregate and sustain information on hospices' quality choices for much longer and with greater accuracy, and can be thought of as helping reputation persist longer. However, the website has seen limited consumer participation and awareness. Figure 3a looks at how the visits-per-patient-day varies as reputation becomes less persistent. The counterfactual is done for two identical firms competing in a market.

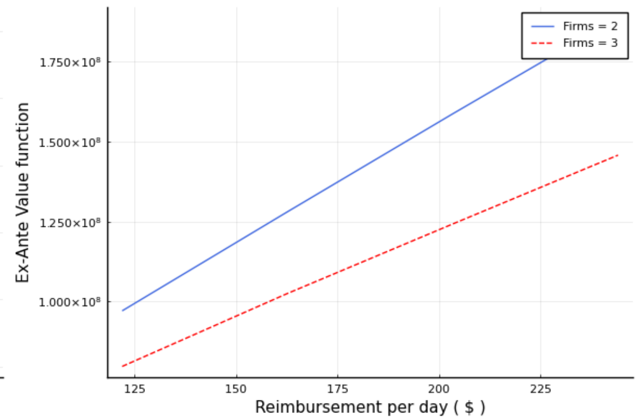
The takeaway is that hospice quality falls as reputation decays faster. The intuition is that the marginal gain from increasing quality is smaller: since reputation is decaying more quickly, an increase in quality leads to a smaller increase in future sales, so it is no longer optimal to incur the higher cost from higher quality.

Given that we saw heterogeneity across markets when it came to visit cost (Table 8), I also study how this graph changes if the hospices were to have lower marginal cost. Hospices choose higher equilibrium quality when costs are lower; however, perhaps surprisingly, I see that hospices react more strongly to a change in the reputation depreciation rate (τ). That is, for lower marginal cost, equilibrium quality drops by more as τ goes from 0.4 to 1.0. This suggests that improving reputation persistence will have heterogeneous effects across markets based on the underlying cost primitives, with lower marginal cost regions seeing stronger quality responses.

Figure 4: Quality choice against different Medicare reimbursement rates.



(a)



(b)

Notes: The figure reports the average quality choice and average ex-ante value function of one hospice over the simulation. The model is solved for two and three identical hospices via policy iteration.

7.3 Increasing Medicare reimbursement rates

To see how hospices react to higher/lower reimbursement rates, my second set of counterfactuals study how hospice quality changes as Medicare rate increases. This is motivated by the fact that hospices (and other healthcare providers) have frequently complained Medicare reimbursement rates are too low and have not kept up with inflation rates.

Figure 4a reports a counterfactual where 2 and 3 identical firms make quality choices under different Medicare reimbursement rates. It can be seen that i) higher Medicare prices lead to higher qualities being chosen by each firm, and ii) quality choice is higher with more firms due to greater competition.

Lower Medicare reimbursement rates can also influence equilibrium quality choice by affecting entry and exit incentives. Figure 4b shows how the average ex-ante value function of a hospice over the simulation changes as Medicare rate increases. Higher (lower) Medicare rate leads to higher (lower) ex-ante value function, which may lead to entry (exit). The change in the number of firms therefore leads to an additional impact on quality. In other words, higher Medicare rates can lead to higher quality both through more intense competition between existing firms as well as through entry of new firms.

7.4 Tying Medicare reimbursement to quality choice

My final set of counterfactuals investigates alternative reimbursement schemes for Medicare. The current reimbursement scheme involves Medicare paying hospices for every day a patient is enrolled, i.e. a per-day per-patient scheme. If we want to achieve a target quality level at the lowest possible Medicare spending, a natural route to investigate is to tie part of the reimbursement to quality.

To that end, I solve the model for hybrid per-day per-visit schemes, with the results shown in Table 10. I progressively reduce the per-day rate and increase the per-visit rate while ensuring that under the new terms, the equilibrium quality choice remains at 0.78 visits-per-patient-day. The total Medicare spending is calculated for each scheme; for clarity, I normalize the spending at each scheme by the total Medicare spending at the per-day per-patient rate (i.e. Medicare's current scheme).

I find that Medicare can reduce per-day rate and raise per-visit rate in such a way that total spending declines even as equilibrium quality choice stays the same. The takeaway is that potential cost savings can be achieved by shifting weight from per-day to per-visit reimbursement. Note that how far I can increase the per-visit rate is limited by the estimated marginal cost, since a per-visit rate exceeding the marginal cost will lead to hospices making unlimited visits to maximize profits.[25]

There is a trade-off associated with this scheme however – as we change the reimbursement structure in this direction, the average ex-ante value function of each hospice decreases, indicating that hospices are making lower profits. As a result, this scheme may risk inducing exit among hospices. Therefore, when implementing this scheme, the policymakers have to be wary of not inducing exit and worsening equilibrium quality.[26]

[25] An alternative presentation of this counterfactual could have been raising spending such that ex-ante value function of firms stay the same, which would mean entry-exit incentives remain unchanged. In that setting, the equilibrium quality would rise, and we can compare the additional spending with the increase in quality. I choose to not do that because a lot of discussion in the US – specifically around hospices – are about how to reduce healthcare spending. This counterfactual suggests a way to do so without sacrificing quality.

[26] I repeat this counterfactual with 3 identical hospices and obtain the same qualitative findings; see Appendix F.

Table 10: Reimbursement schemes that lead to 0.78 visits-per-patient-day in a duopoly.

Per-day rate	**Per-visit rate**	**Medicare Spending (normalized)**	**Ex-Ante Value Function (normalized)**
223.7	0.0	1.0	1.0
203.4	20.0	0.97	0.93
183.1	40.0	0.96	0.86
142.4	80.0	0.91	0.71
101.7	120.0	0.87	0.57
61.0	160.0	0.83	0.43

Notes: The rates are in US dollars. The total Medicare spending associated with each scheme is normalized with respect to the spending of the per-day scheme. The average ex-ante value function is normalized with respect to the ex-ante value function of the per-day scheme. As we move down rows, equilibrium quality stays the same yet Medicare spending declines.

Finally, instead of thinking of this counterfactual as tying hospice reimbursement to visits specifically, we can also interpret it as tying reimbursement to some measure of hospice quality. This measure of quality could be obtained via a comprehensive hospice-quality index constructed by CMS or through Family Caregiver surveys conducted every few weeks.

7.5 Discussion

The qualitative results of the last two counterfactuals – Medicare reimbursement increase leads to higher equilibrium quality, and tying reimbursement partially to quality allows us to maintain the same equilibrium quality with lower spending – hold even with a simpler model of quality competition between hospices. That is, if $\beta = 0$ so that hospices only chose quality taking into account its effect on current market share (i.e. competing in quality), then the qualitative results from the counterfactuals still hold. So why include reputation in my model? This is because i) it represents an accurate modeling of provider competition, ii) it allows us to recover a more accurate estimate of the cost function, and iii) it shows the promise of a review website such as Hospice Compare in improving hospice quality.

A possible issue with subsidizing visits is that it could encourage spurious visits or fraudulent reporting by hospices. However, I believe that is less of a concern for the following reasons:

1. As mentioned before, we can interpret the counterfactual as showing we should tie reimbursement to some measure of quality, not necessarily visits.

2. My estimates suggests that reputation effects are salient in this industry. Thus, a hospice that makes spurious visits can develop a bad reputation and lose future consumers, so reputation effects might rein in such issues.

3. Medicare already face issues like fraudulent reporting and upcoding in other settings. Therefore, to discourage such actions, Medicare can deploy the sort of documentation and monitoring that it already employs. For example, i) it can adopt finer contract structures (such as reimbursing hospices for up to 3 visits a week), ii) it can enforce stronger monitoring and auditing, for instance the patient's family may be asked to sign off after every visit.

Finally, another possible advantage of the hybrid reimbursement scheme is that it may reduce incentives for hospices to avoid patients with short lengths-of-stay who require many visits (Gruber et al. (2023)).

8 Conclusion

Using firm-level data from California, I study quality choice by hospices, uncover the importance of hospice reputation for consumers, and explore counterfactual policies that can incentivize higher hospice quality and lower Medicare spending. I build and estimate a structural model of consumer demand and hospice quality choice to do so. My measure of hospice quality is the average number of visits made by a hospice to its patients per day in a year, and I define reputation of a hospice to be a function of its current and past quality choices. As a result, a hospice can accumulate reputation over time by consistently choosing high quality. I use my structural model to quantify the importance of reputation for consumers choosing hospices and estimate the hospice cost function. Reputation plays an important role in consumer demand for hospices, and past reputation decays at an annual rate of 46%. The cost of an additional visit by a hospice to a patient is approximately

$192. Counterfactuals show that hospice quality increases with higher Medicare prices and greater persistence of reputation. I compare the current per-day Medicare reimbursement scheme with a hybrid per-day per-visit scheme, and find that the latter can incentivize the same level of hospice quality at lower Medicare spending.

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Online Appendix

A Additional data preparation for structural estimation

Censoring quality variable: To deal with high outlier values of the quality choice variable, I censor the visits-per-patient-day variable at the 90th percentile. That is, for quality values which are greater than the 90th percentile, I recode those values to be equal to the 90th percentile value.

Missing observations and fixing Firm IDs: I perform some additional data-cleaning before doing the analysis. First, for 24 hospices, it seems from looking at the data that a hospice forgot to respond in a certain year. This conjecture is due to the fact that the hospice was operating and serving a large number of patients in both the past few and next few years, but suddenly had its observation missing in one year. These non-responses are scattered across several years, with most in 2003 and only one after 2010. For these instances, I copy over the hospice's previous year's sales and characteristics into the missing year. This comprises 24 imputed data points in my dataset of approximately 2000 observations.

The data provider HCAI assigns an ID to each hospice, called the OSHPD ID. For 3 OSHPD IDs, they disappear from the data for 3-4 years before returning. I classify these as new entries by different entities.

Finally, for a few hospices, their OSHPD IDs change despite everything else about them staying the same (their name, location, etc.). For these hospices, I harmonize each to have the same OSHPD ID throughout.

I am happy to supply the code file for verification.

Imputing reputation stock of older hospices: For hospices which enter after 2002 (earliest period in my dataset), I can construct their exact reputation stock because I set their reputation during entry to be 0. However, a number of hospices are present from before the year 2002, and for them, I do not observe their pre-2002 quality choices.

While this does not affect my demand estimation (observe that the Berry (1994) allows me to difference out log of outside option share to account for rivals' characteristics), it will prevent me from conducting my supply side estimation. This is because a hospice's quality choice in each observed state is influenced by all its rivals' reputation stocks.

As a result, for the year 2002, I impute the reputation stock of hospices arriving before 2002 using a flexible functional form. First, for all hospices that arrived *after* 2002, I regress their reputation stocks on a large number of observed covariates. I use these regression coefficients to impute the reputation stocks – for the year 2002 – of all hospices that arrived before 2002. Now that I know their reputation stock coming into year 2002, I can use all their successive quality choices to update their reputation stock in future periods.

B Robustness checks for demand estimation

I conduct additional robustness checks for demand estimation. First, I reestimate demand without imposing the AR(1) structure on ξ_{jt} and continue finding evidence of reputation; see Table 13. Second, instead of BLP-IVs, I redo my demand estimation with Gandhi-Houde IVs and obtain similar results. I also use two additional instruments – i) measures of reported average cost of a visit for each hospice from the cost data, and ii) Medicare prices. The average reported visit cost acts as a cost shifter. Medicare prices impact quality choice (see counterfactuals) while being orthogonal to firm-specific demand shocks since they are set at the national level and vary by county based on Medicare cost indices. I also repeat my demand estimation without the county-dummy-by-fuel-price IVs. All specifications yield similar results; see Tables 11 and 12 for these and other robustness checks. In particular, I use different subsets of the aforementioned instruments to ensure my results aren't driven by any specific instrument. Finally, a previous version of the paper used "visits-per-patient" as the quality choice variable and obtained similar results.

B.1 With persistence of ξ

I estimate the following system of equations using 2-step GMM.

$$ln(s_{jt}) - ln(s_{0t}) = \alpha_{m(j)} + X'_{jt}\beta_d + \sigma_n ln(s_{j|gt}) + \xi_{jt} + \\ \eta[\mathrm{a}_{jt} + (1-\tau)\mathrm{a}_{jt-1} + (1-\tau)^2\mathrm{a}_{jt-2} + \dots]$$

$$\xi_{jt} = \rho\xi_{jt-1} + \varepsilon_{jt}$$

Here τ denotes depreciation rate of reputation, ρ denotes persistence of unobserved hospice-specific demand shocks, σ_n is the nest parameter, and η is the impact of quality choice on the reputation stock.

For robustness, I re-estimate this model using different sets of instruments. I continue to find evidence of reputation and its persistence.

Also note that the absence of reputation persistence would mean $\tau = 1$; therefore the t-stat for τ is $\frac{\hat{\tau}-1}{s.e.(\tau)}$.

Table 11: Results of demand estimation with BLP-IVs.

	(1)	(2)	(3)
τ	0.345	0.460	0.142
	(0.145)	(0.143)	(0.111)
ρ	0.767	0.478	0.444
	(0.066)	(0.150)	(0.164)
σ	0.601	0.584	0.685
	(0.056)	(0.063)	(0.078)
η	0.400	0.626	0.230
	(0.106)	(0.191)	(0.117)
Hospice inpatient unit	0.071	0.113	0.142
	(0.168)	(0.080)	(0.081)
Pediatric program	0.241	0.282	0.212
	(0.084)	(0.086)	(0.076)
Bereavement services	-0.013	0.010	-0.025
	(0.045)	(0.038)	(0.033)
Day care for adults	0.032	0.112	-0.066
	(0.159)	(0.205)	(0.123)
For-profit	-0.164	-0.216	-0.035
	(0.102)	(0.083)	(0.114)
Agency type: free-standing	-0.222	-0.191	-0.176
	(0.152)	(0.114)	(0.091)
Agency type: home health based	-0.316	-0.238	-0.166
	(0.201)	(0.150)	(0.139)
Agency type: hospital-based	-0.277	-0.205	-0.182
	(0.163)	(0.131)	(0.121)

Notes: (1): BLP-IV with county-by-fuel IVs and log-slope-per-visit and inflation-adjusted Medicare prices; (2): BLP-IV with log-slope-per-visit and inflation-adjusted Medicare prices; (3): BLP-IV with log-slope-per-visit. The demand model is estimated using 2-step nonlinear GMM. Robust standard errors are in parentheses. Estimates of county fixed effects are suppressed.

Table 12: Results of demand estimation with GHIVs.

	(1)	(2)	(3)	(4)
τ	0.270	0.097	0.256	0.329
	(0.136)	(0.062)	(0.136)	(0.160)
ρ	0.886	0.861	0.893	0.629
	(0.098)	(0.063)	(0.070)	(0.152)
σ	0.578	0.685	0.630	0.581
	(0.085)	(0.082)	(0.075)	(0.116)
η	0.319	0.164	0.274	0.414
	(0.149)	(0.074)	(0.080)	(0.206)
Hospice inpatient unit	0.102	0.094	0.094	0.124
	(0.185)	(0.158)	(0.173)	(0.103)
Pediatric program	0.238	0.146	0.227	0.296
	(0.110)	(0.099)	(0.098)	(0.091)
Bereavement services	0.003	-0.034	-0.009	-0.008
	(0.056)	(0.042)	(0.051)	(0.047)
Day care for adults	-0.043	-0.083	-0.032	-0.078
	(0.218)	(0.108)	(0.115)	(0.299)
For-profit	-0.179	-0.027	-0.134	-0.200
	(0.239)	(0.168)	(0.203)	(0.120)
Agency type: free-standing	-0.217	-0.200	-0.205	-0.169
	(0.273)	(0.162)	(0.222)	(0.115)
Agency type: home health based	-0.358	-0.255	-0.306	-0.220
	(0.277)	(0.186)	(0.250)	(0.166)
Agency type: hospital-based	-0.318	-0.244	-0.275	-0.222
	(0.260)	(0.178)	(0.257)	(0.143)

Notes: In what follows, GHIV stands for Gandhi-Houde IVs (Gandhi and Houde (2019)). (1): GHIV with county-fuel IVs (2): GHIV with county-fuel IVs and log-slope-per-visit; (3): GHIV with county-fuel IVs and log-slope-per-visit and inflation-adjusted Medicare prices; (4): GHIV with log-slope-per-visit and inflation-adjusted Medicare prices. The demand model is estimated using 2-step nonlinear GMM. Robust standard errors are in parentheses. Estimates of county fixed effects are suppressed.

B.2 Without imposing persistence of ξ

I estimate the following system of equations using 2-step GMM.

$$ln(s_{jt}) - ln(s_{0t}) = \alpha_{m(j)} + X'_{jt}\beta_d + \sigma_n ln(s_{j|gt}) + \xi_{jt} +$$
$$\eta[\mathrm{a}_{jt} + (1-\tau)\mathrm{a}_{jt-1} + (1-\tau)^2\mathrm{a}_{jt-2} + \ldots]$$

I no longer impose the autoregressive structure on ξ_{jt}. This means I now interact the instruments with the ξ_{jt} term directly, rather than the innovation of ξ_{jt} as in the previous subsection.

Here τ denotes depreciation rate of reputation, σ_n is the nest parameter, and η is the impact of quality choice on the reputation stock.

For robustness, I re-estimate this model using different sets of instruments. I continue to find evidence of reputation and its persistence.

Also note that the absence of reputation persistence would mean τ = 1; therefore the t-stat for τ is $\frac{\hat{\tau}-1}{s.e.(\tau)}$.

Table 13: Results of demand estimation without imposing persistence of ξ.

	(1)	(2)	(3)	(4)
τ	0.105	0.593	0.611	0.390
	(0.099)	(0.155)	(0.116)	(0.185)
σ	0.645	0.572	0.574	0.549
	(0.039)	(0.052)	(0.048)	(0.071)
η	0.104	1.021	1.090	0.837
	(0.053)	(0.357)	(0.192)	(0.466)
Hospice inpatient unit	0.143	0.114	0.115	0.165
	(0.072)	(0.071)	(0.070)	(0.095)
Pediatric program	0.276	0.290	0.285	0.240
	(0.053)	(0.104)	(0.086)	(0.159)
Bereavement services	0.011	4.223e-04	-1.910e-04	-0.050
	(0.033)	(0.047)	(0.056)	(0.075)
Day care for adults	-0.066	0.258	0.281	0.072
	(0.264)	(0.346)	(0.373)	(0.291)
For-profit	-0.069	-0.275	-0.282	-0.285
	(0.047)	(0.071)	(0.075)	(0.133)
Agency type: free-standing	-0.083	-0.122	-0.114	-0.211
	(0.077)	(0.139)	(0.129)	(0.154)
Agency type: home health based	-0.191	-0.157	-0.144	-0.193
	(0.085)	(0.188)	(0.158)	(0.192)
Agency type: hospital-based	-0.202	-0.103	-0.086	-0.172
	(0.092)	(0.182)	(0.156)	(0.171)

Notes: (1): BLP-IVs with county-fuel IVs (2): BLP-IVs; (3): BLP-IVs and log-slope-per-visit; (4): sum of rivals, sum of for-profits, and log-slope-per-visit. The demand model is estimated using 2-step nonlinear GMM. Robust standard errors are in parentheses. Estimates of county fixed effects are suppressed.

C Additional descriptive statistics and regression results

Table 14: Regressing estimated ξ_{jt} on quality.

	ξ_{jt}			
	(1)	(2)	(3)	(4)
Intercept	0.069			
	(0.073)			
Quality	-0.089	-0.096	-0.122	-0.132
	(0.101)	(0.108)	(0.111)	(0.126)
Count of rivals			0.013	0.015
			(0.012)	(0.012)
County Fixed Effects		Yes	Yes	Yes
Controls				Yes
Obs.	1,002	1,002	1,002	1,002

Notes: ξ_{jt} is recovered as part of the demand estimation. Controls are hospice inpatient unit indicator, pediatric program indicator, bereavement for non-hospice-survivors indicator, day care for adult services indicator, for-profit status, and agency types indicators. Standard errors clustered at county-level in parentheses.

Table 15: Regressing market share on age of hospices.

	log(market share)
Age	0.344***
	(0.030)
Age (squared)	-0.016***
	(0.002)
County Fixed Effects	Yes
Obs.	1,002

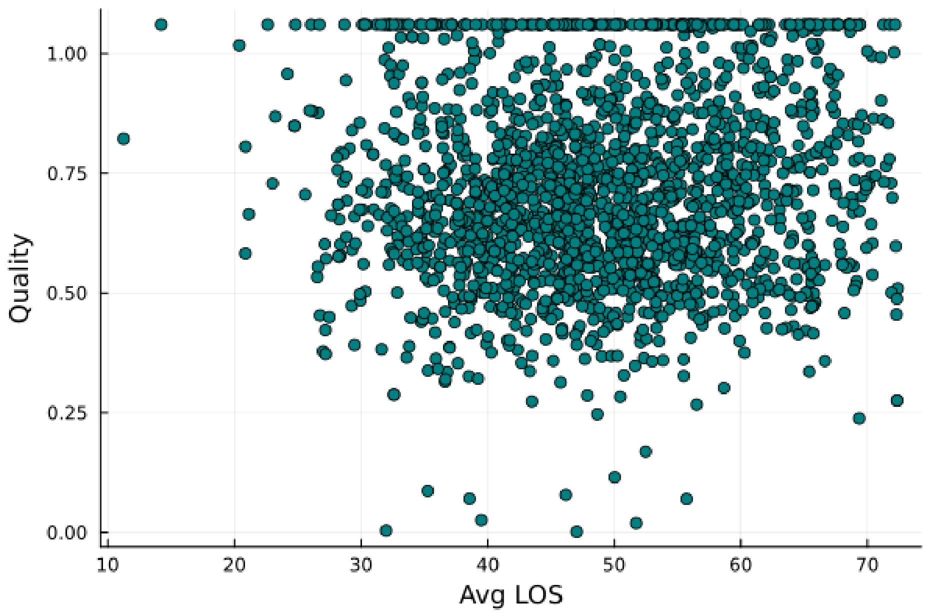
Notes: Standard errors clustered at county-level in parentheses.

Table 16: Regressing quality on average lengths-of-stay.

	Visits-per-patient-day		
	(1)	(2)	(3)
Count of rivals	0.014***	0.009**	0.009**
	(0.002)	(0.003)	(0.003)
Avg LOS	-0.0003	-0.001	-0.005
	(0.001)	(0.001)	(0.004)
Avg LOS squared			4.2×10^{-5}
			(3.9×10^{-5})
County Fixed Effects	Yes	Yes	Yes
Controls		Yes	Yes
Obs.	1,956	1,956	1,956

Notes: This regression drops observations of average lengths-of-stay above the 95th percentile (approximately 71+ days) to prevent outliers affecting the results. Standard errors clustered at county-level in parentheses. "Avg LOS" stands for average length-of-stay. Controls are hospice inpatient unit indicator, pediatric program indicator, bereavement for non-hospice-survivors indicator, day care for adult services indicator, for-profit status, and agency types indicators.

Figure 5: Scatterplot of average length-of-stay and quality.



Notes: “Avg LOS” stands for average length-of-stay.

C.1 Length of stay

Length of stay	% of patients
0-7 days	30.89
8-30 days	29.38
31-90 days	21.7
91-179 days	9.52
180+ days	8.51

Table 17: Distribution of patient length-of-stay.

In my dataset, I observe the number of patients in a hospice who stayed for i) 0-7 days, ii) 8-30 days, iii) 31-90 days, iv) 91-179 days, v) 180+ days. Table 17 uses this data to construct the total number of patients in my dataset within each LOS bracket. Over 80% of the patients stay for less than 90 days.

Table 18 shows the distribution of people within each LOS-bracket by hospice-year. For nearly all hospices, most of their patients stay for less than 90 days.

	10%	25%	50%	75%	90%
0-7 days	19.76	24.66	30.3	36.31	41.81
8-30 days	21.93	25.05	29.29	33.01	36.33
31-90 days	16.15	18.9	21.7	25.17	29.44
91-179 days	5.45	7.47	9.41	11.43	13.92
180 days	2.24	4.48	7.2	10.73	14.52

Table 18: Distribution of shares of lengths-of-stay of patients. Each observation is at the hospice-year level.

C.2 Disease category

Diagnosis	% of patients
Cancer	39.6
Heart	10.27
Dementia	13.57
Lung	6.39
Kidney	2.49
Brain stroke	4.47

Table 19: Percentage of hospice patients by diagnosis.

The dataset classifies the patients at a hospice by broad catergories of diagnosis. This allows me to see which diagnosis are more prevalent and whether hospices specialize in certain diagnosis over others. Table 19 shows that most hospice patients suffer from cancer; the second and third largest diagnosis categories are heart disease and dementia.

To see if hospices are specializing by diagnosis, or whether patients select into hospices by their diagnosis, Table 20 shows the distribution of patients of each diagnosis by hospice-year. This shows that most of a hospice's patients are suffering from cancer, heart disease or dementia, and there is no clear sign that a hospice gets most of its patients from one disease category alone.

	10%	25%	50%	75%	90%
Cancer	24.23	31.25	39.63	49.42	60.0
Non cancer	40.0	50.52	60.36	68.75	75.73
Heart disease	3.91	6.67	9.81	13.29	17.82
Dementia	3.33	6.5	11.72	18.03	25.0
Lung disease	2.61	4.27	6.18	8.46	11.12
Kidney	0.0	1.34	2.32	3.49	5.21
Liver	0.0	0.98	1.85	2.91	4.3
Brain stroke	0.0	1.53	3.48	5.89	8.96
HIV	0.0	0.0	0.0	0.13	0.46
Coma	0.0	0.0	0.0	0.0	0.35
Diabetes	0.0	0.0	0.0	0.0	0.44
ALS	0.0	0.0	0.26	0.61	1.03

Table 20: Distribution of shares of patients at a hospice per disease category. Each observation is at the hospice-year level.

C.3 Care type

	10%	25%	50%	75%	90%
Routine care	98.72	99.46	99.8	99.94	100.0
Inpatient care	0.0	0.0	0.04	0.23	0.69
Respite care	0.0	0.0	0.06	0.16	0.36
Continuous care	0.0	0.0	0.0	0.01	0.08

Table 21: Distribution of days of service provided for each hospice-year per care-type.

Medicare reimburses hospices with rates based on whether it provided routine care, inpatient care, respite care, and continuous care. Routine care has the lowest Medicare reimbursement rate, while other care types are much higher (see Figure 8).

The dataset divides the total days of care provided by a hospice into days providing each type of care. I calculate the fraction of care days in my dataset for each type of care. Table 21 shows that nearly all of a hospice's care days are for routine care.

	10%	25%	50%	75%	90%
Registered nurses	8.82	11.42	14.44	18.21	22.48
Licensed vocational nurses	0.0	0.0	0.64	4.43	8.9
Homemakers	5.39	7.69	11.24	16.85	24.21
Physicians	0.0	0.0	0.01	0.25	0.77
Social service workers	1.74	2.59	3.55	4.73	6.05
Chaplains	0.27	1.02	1.84	2.87	4.08

Table 22: Distribution of visits by staff type for each hospice-year. The unit is visit per patient.

	Min	10%	25%	50%	75%	90%	Max
Share	0.0	0.0	0.0	0.00546448	0.0136986	0.0291228	0.590909

Table 23: Distribution of share of patients discharged to other hospices for each hospice-year.

C.4 Market definition and selection

C.4.1 Defining markets to be at the county level

While I define my market to be at the level of a county, some papers on healthcare providers define markets at a larger level. Furthermore, some hospices advertise as being available for service in multiple counties at once.

In my setting, I can rule out larger market definitions using my dataset. For each hospice, the dataset includes measures of the fraction of patients that arrive from each county. This allows me to isolate the fraction of a hospice's patients coming from the "home county" of the hospice (i.e. the county where the hospice is located) versus the fraction that is coming from an "away county".

Variable	5%	10%	25%	50%	75%	90%
Home County Share	0.39	0.54	0.74	0.91	0.99	1.0

Table 24: Distribution of share of patients coming from home county. Observations are at the hospice-year level.

As Table 24 make clear, the vast majority of a hospice's patients arrive from the county where it is located. As a result, I define the relevant market to be at the county level.

For the hospices which are below the reported 5th percentile, I see which counties they are located in. This is to check if there is one specific county where nearly all hospice patients are from outside that county. That turns out not to be the case – the hospices with home county share below the 5th percentile are scattered across over 10 counties.

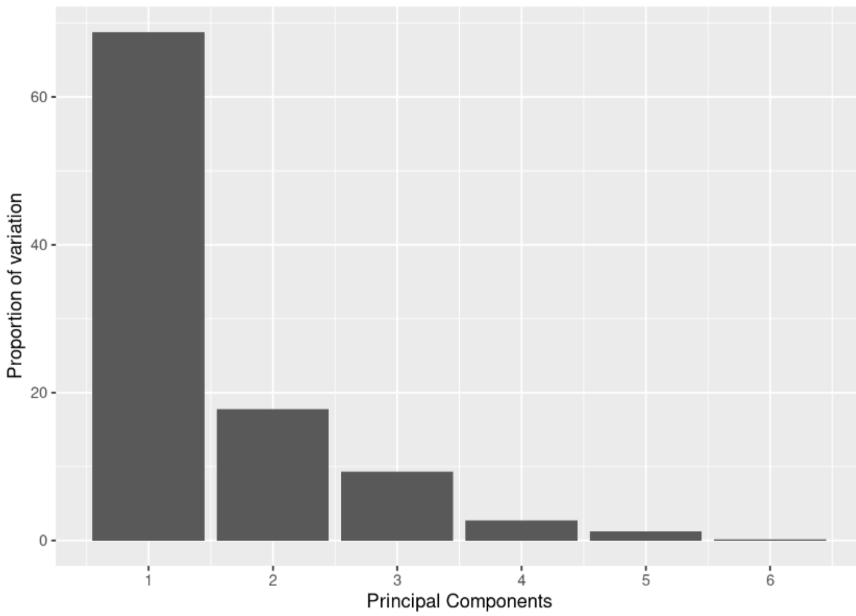
C.4.2 Market selection

I focus on the following 39 counties in California: Alameda, Contra Costa, Amador, Butte, San Joaquin, El Dorado, Fresno, Humboldt, Del Norte, Imperial, Kern, Kings, Lake, Madera, Marin, Mariposa, Mendocino, Merced, Monterey, Napa, Nevada, Placer, Sacramento, San Francisco, San Luis Obispo, San Mateo, Santa Barbara, Santa Clara, Santa Cruz, Shasta, Siskiyou, Solano, Sonoma, Stanislaus, Sutter, Tehama, Tulare, Tuolumne, Yolo.

I drop counties which saw very high number of hospices by the end of the dataset's coverage (2018). The counties which are dropped for this purpose are (with their number of hospices in 2018 in parentheses) are: Los Angeles (377), Orange (63), Riverside (34), San Bernardino (92), San Diego (31), and Ventura (40).

The highest number of hospices for a county in my dataset is 23 (Alameda). Therefore, I am implictly dropping counties which had more than 30 hospices by the end of the dataset's coverage.

C.5 Principal components analysis of visits by staff type



	PC1	PC2	PC3	PC4	PC5	PC6
rnpat	0.4223153	0.7939506	-0.3565230	0.2495343	0.0433376	0.0061776
physnpat	0.0151012	-0.0068927	-0.0221099	-0.0032503	-0.0416916	-0.9987426
socsvcspat	0.1317572	0.1580743	-0.1338480	-0.9503556	0.1912078	-0.0010246
lvnpat	0.1834381	-0.5006202	-0.8411163	0.0710564	0.0519645	0.0224485
homemkrpat	0.8730719	-0.3053850	0.3769924	0.0254161	0.0411040	0.0051643
chaplainpat	0.0903201	0.0269610	-0.0699097	-0.1698567	-0.9774627	0.0440833

Figure 6: Principal components analysis of visits by staff type. The staff types are (in order of presentation) Registered Nurse, Physician, Social service worker, Licensed Vocational nurse, Home-maker, and Chaplains. The PCA analysis shows that the first Principal Component explains most of the variation in visits by a hospice.

D Discussion on capacity constraints

Table 25: Regression of total visits by a hospice in a year on covariates.

	Total visits		log(total visits)	
	(1)	(2)	(3)	(4)
Patients	53.992***	44.326***		
	(3.766)	(7.030)		
Count of rivals	2186.478***	1929.452***		
	(489.214)	(503.053)		
Count of rivals squared	-62.318***	-52.588**		
	(15.967)	(17.719)		
log(patients)			1.104***	1.059***
			(0.016)	(0.038)
log(total firms)			0.252**	0.193**
			(0.072)	(0.062)
County Fixed Effects	Yes		Yes	
Year Fixed Effects	Yes	Yes	Yes	Yes
Firm Fixed Effects		Yes		Yes
Controls	Yes	Yes	Yes	Yes
Obs.	2,060	2,034	2,060	2,034

Notes: Standard errors clustered at county-level in parentheses. Controls are hospice inpatient unit indicator, pediatric program indicator, bereavement for non-hospice-survivors indicator, day care for adult services indicator, for-profit status, and agency types indicators.

There may be some concern about whether capacity constraints are present in this industry. That is, each hospice really has a fixed number of staff over time, and those staff get spread out over however many patients the hospice has in any given year.

I claim that capacity constraints are not a concern in my setting. This is for the following reasons:

1. **Market share is positively correlated with visits-per-patient-day:** If capacity constraint was present, it would create negative correlation between visits-per-patient-day and market share. The reason is that as a hospice gets more patients, it can make fewer visits to each of

them due to the same number of staff getting spread out more thinly across a larger patient pool. However, the pattern I see in my data is the opposite. I find that market share is *positively* correlated with visits-per-patient-day; see Table 5. This suggests that hospices with more patients also make more visits per patient, which is consistent with the idea that they are not capacity constrained.

2. **Firms increase the total visits they make when facing greater competition:** After controlling for total patients, I see total number of visits by a hospice increases as there are more competitors. The results are reported in Table 25. This suggests that, holding fixed the patient volume, a hospice makes more visits when facing more rivals in the market. This goes against the idea that hospices have a fixed number of staff as competitors enter and that these staff are distributed over whatever number of patients the hospice has. Instead, it suggests that hospices respond to entry by making more visits in total.

3. **Institutional details suggest capacity constraint is unlikely to be an issue:** Hospices are not really limited by physical space since they generally serve patients at patient residences. From my conversations with a hospice admin and from online readings, it seems that hospices hire nurses living around the area where their patients reside. The hospice nurses often drive directly from their home to various patient residences. Therefore, there isn't a capacity constraint with hospices like there is with nursing homes or hospitals.

E Empirical policy function

I regress the quality measure visits-per-patient-day on a variety of covariates to see how quality choice is affected by state variables. This sheds light on how hospices react to reputation levels and product characteristics of themselves and rivals. I interpret this regression as reflecting the policy function used by hospices when making quality decisions.[27]

I use the demand estimates from the previous section to construct reputation stock of each firm in every market-year, the attractiveness of their product attributes ($X'_{jt}\beta_d$), and their hospice-specific demand unobservables (ξ_{jt}). In a dynamic game, these values of a hospice and its rivals would enter the policy function.

To account for competitive effects from rivals, I include the following state variables for a given hospice: number of rivals, difference between average reputation of rivals and own, difference between average $X'_{jt}\beta_d$ of rivals and own, and difference between average ξ_{jt} of rivals and own.

I approximate the empirical policy function using two different methods: a 2SLS regression and a flexible regression using third-order polynomials. Below I explain the exact implementation of both methods.

Both specifications for estimating policy function yield very similar results for the slope of the cost function. I generally find that "count of rivals" is the key driver of the cost estimation results, and that is preserved with both the 2SLS and other flexible functions I have tried.

E.1 Policy function using 2SLS

When recovering the empirical visit-choice policy function from the data, I try to be cognizant of the issues raised in Berry and Compiani (2023) and Berry and Compiani (2021). Briefly, a regression of visit choice on dynamic states such as reputation stock can suffer from endogeneity

[27]Note that for-profit status changes over time for same hospice, allowing me to identify the for-profit coefficient even with hospice fixed effects.

problem in the presence of serially correlated unobservables.[28] To deal with such issues, I take the following steps. First, I include firm-specific fixed effects in my regression. Second, following the suggestions from Berry and Compiani (2023), I use instruments to correct the endogeneity problem. I use lagged $X_{jt}'\beta_d$ and lagged ξ_{jt} to instrument for the dynamic state (difference between average reputation of rivals and own). These instruments are valid because current period's quality choice is affected by *current* $X_{jt}'\beta_d$ and ξ_{jt}, while the last period's reputation stock is affected by last period's $X_{jt}'\beta_d$ and ξ_{jt}. (Note that current period's $X_{jt}'\beta_d$ and ξ_{jt} are included in the regression as well.)[29]

The resulting 2SLS regression has signs on the state variables that match the general shape of the policy functions I observe in my counterfactuals.[30] Furthermore, I check the 2SLS first-stage regression of dynamic states on the exogenous variables and instruments, and find them to have signs predicted by economic theory; see Appendix E.1.1 for results and discussion. The results of the regression are given in Table 26. To summarize, hospices choose higher quality when facing more rivals, and when facing rivals with better reputation, $X_{jt}'\beta_d$, and ξ_{jt}. They choose lower quality when they themselves have higher reputation, $X_{jt}'\beta_d$, and ξ_{jt}.

While I use the first reported regression in Table 26 as my estimate of empirical policy function, I also conduct robustness checks using the hospice-reported costs. In my dynamic oligopoly

[28]For instance, if the unobserved shocks are serially correlated and also affect the visit choice, then the dynamic state would be influenced by past shocks; past shocks are correlated with current shocks; and so the current period dynamic state is correlated with the current shock.

[29]Berry and Compiani (2023) go further and suggest using nonparametric IV methods such as GIV, with their argument being that the policy function of the firm may be highly nonlinear. While I run a linear regression to simplify estimation, I conduct the following check. In my counterfactual, I solve for the policy function for heterogeneous hospices. Then, I check the sign on the policy function for different values of the state variables (own reputation, rival reputation, own and rival $X_{jt}'\beta_d$, own and rival ξ_{jt}). I find that the signs on the solved policy function match the signs on the regression, i.e. they agree on which direction each state variable moves the quality choice. Thus, while the *exact* policy function may be nonlinear, I argue that this regression represents a meaningful approximation of it.

[30]In additional counterfactuals (not reported in the main text, available upon request), I solve for the policy function for heterogeneous hospices. I find that, holding other state variables constant, a hospice chooses higher quality if its rivals have high reputation/$X_{jt}'\beta_d/\xi_{jt}$, and lower quality if the hospice itself has high reputation/$X_{jt}'\beta_d/\xi_{jt}$.

estimation, I will estimate the "visit cost" rather than use the hospice-reported costs.[31]

Note that while I use this empirical policy function for the estimation, I do not use it for my counterfactuals; instead, I will numerically solve for the optimal policy function.

Table 26: Estimates from 2SLS regression of visit-per-patient-day on covariates.

	Visits-per-patient-day		
	(1)	(2)	(3)
Count of rivals	0.041***	0.044***	0.045***
	(0.011)	(0.012)	(0.012)
For-profit	0.124*	0.035	0.036
	(0.061)	(0.061)	(0.061)
Fuel price	-0.042	-0.037	-0.040
	(0.022)	(0.025)	(0.025)
Medicare rate	0.019*	0.027**	0.028**
	(0.008)	(0.010)	(0.010)
Average $X'_{jt}\beta_d$ of rivals - own $X'_{jt}\beta_d$	0.401**	0.371**	0.368**
	(0.151)	(0.141)	(0.141)
Average ξ_{jt} of rivals - own ξ_{jt}	0.455***	0.423***	0.430***
	(0.094)	(0.082)	(0.085)
Average reputation of rivals - own reputation (lagged)	1.589**	1.381**	1.379**
	(0.542)	(0.470)	(0.472)
Visit cost		-0.011	-0.011
		(0.007)	(0.007)
Intercept cost			-0.021
			(0.019)
Hospice Fixed Effects	Yes	Yes	Yes
Obs.	1,608	1,130	1,130

Notes: Estimates of hospice-specific fixed effects suppressed. The endogenous variable in the regression is the dynamic state "Average reputation of rivals - own reputation (lagged)". The instruments used are ξ_{jt-1} and $X'_{jt-1}\beta_d$. The visit cost and intercept cost used in this regression are those reported by the hospices in the cost data.

The first stage of Bajari et al. (2007) requires an estimate of the policy function from the data.

[31] One concern is that the Medicare reimbursement rate partly reflects cost differences across regions; thus, an increase in Medicare reimbursement rate does not necessarily mean an increase in profits per patient. This means the interpretation of the coefficient on the Medicare reimbursement rate is less clear. I have two arguments in response. First, I include firm-specific measures of cost in the empirical policy function and still find similar coefficients on the Medicare reimbursement rate. Second, the empirical policy function is used only to help estimate the cost function and does not play a role in my counterfactuals. In my counterfactuals, I will numerically solve for the policy function rather than rely on the empirical policy function.

To this end, I use a variant of the empirical policy function in Table 26. I make the following adjustments from the empirical policy function. First, recall that I discretize visit choice into 6 tiers; as a result, I use the empirical policy function and random draws from the residual distribution to make predictions of quality choice given values of state variables, then round it to the nearest quality tier. Second, I do not make use of the firm-specific fixed effects for making predictions. Most firms are observed for a few periods, and so the fixed effect estimates are likely to be biased. Also, as alluded to earlier, the fixed effects are identified off of an assumption of linear policy function, which is unlikely to hold in practice.

E.1.1 First stage of 2SLS

In this subsection I dig into the 2SLS regression reported in Table 26. The endogenous variable in that regression was the dynamic state "Average reputation of rivals - own reputation (lagged)". The instruments used were ξ_{jt-1} and $X'_{jt-1}\beta_d$.

Below I disentangle the first stage regression for the 2SLS. Fundamentally, I want to make sure that the instruments are moving the dynamic state in accordance with economic theory. Briefly, a higher ξ_j or $X_j\beta_d$ should cause hospice j to choose lower quality and therefore get lower reputation. On the other hand the rivals would choose higher quality and therefore improve their reputation. These are insights I get from conducting counterfactuals involving heterogeneous hospices.

I therefore run the first-stage IV regression separately on two variables: Average reputation of rivals (lagged) and own reputation stock (lagged). The tables below show the regression of each variable on the exogenous variables and instruments. They confirm that the instruments move the dynamic states in the direction I expect.

Table 27: Estimates from first stage of 2SLS regression.

	Own reputation (lagged)
Count of rivals	0.014***
	(0.002)
For-profit	0.061**
	(0.023)
Fuel price	0.017
	(0.009)
Medicare rate	2.88×10^{-5}***
	(2.372×10^{-6})
Average $X'_{jt}\beta_d$ of rivals - own $X'_{jt}\beta_d$	0.006
	(0.018)
Average ξ_{jt} of rivals - own ξ_{jt}	0.084***
	(0.016)
ξ_{jt-1}	-0.035**
	(0.011)
$X'_{jt-1}\beta_d$	-0.076
	(0.055)
Firm Fixed Effects	Yes
Obs.	1,608

Notes: Estimates of hospice-specific fixed effects suppressed.

Table 28: Estimates from first stage of 2SLS regression.

	Average reputation of rivals (lagged)
Count of rivals	-0.005**
	(0.002)
For-profit	0.010
	(0.018)
Fuel price	0.024**
	(0.007)
Medicare rate	1.91×10^{-5}***
	(1.87×10^{-6})
Average $X'_{jt}\beta_d$ of rivals - own $X'_{jt}\beta_d$	-0.253***
	(0.014)
Average ξ_{jt} of rivals - own ξ_{jt}	-0.062***
	(0.013)
ξ_{jt-1}	0.009
	(0.009)
$X'_{jt-1}\beta_d$	0.026
	(0.043)
Firm Fixed Effects	Yes
Obs.	1,608

Notes: Estimates of hospice-specific fixed effects suppressed.

E.2 Flexible policy function

As an alternative, I use a flexible function of the same state variables to approximate the policy function. I regress quality on county fixed effects and third-degree polynomials of most of the covariates. There are several advantages to this as well. First, it better captures any nonlinearities that the policy function may exhibit and therefore avoids functional form misspecification issues. Second, it allows me to avoid having to put in hospice-specific fixed effects, which I leave out for my cost estimation simulations and may be biased in any case. The disadvantage is I cannot control for serially correlated unobservables by using instrumental variables.

Table 29: Estimates from flexible regression.

	Quality
Count of rivals	0.031**
	(0.010)
Count of rivals squared	-0.002*
	(0.001)
Count of rivals cubed	4.8×10^{-5}
	(3.0×10^{-5})
For-profit	0.116***
	(0.013)
Fuel price	-0.027*
	(0.011)
Medicare rate	-0.002
	(0.003)
County Fixed Effects	Yes
Controls	Yes
Obs.	1,795

Notes: Estimates of county fixed effects suppressed. Controls are third-degree polynomials of the following variables: Average $X'_{jt}\beta_d$ of rivals - own $X'_{jt}\beta_d$, Average ξ_{jt} of rivals - own ξ_{jt}, and Average reputation of rivals - own reputation (lagged).

E.3 Entry policy function

Table 30: Ordered logit of entry count on market characteristics, used as entry transition function.

	Entry count
Firm count	-0.300
	(0.117)
Market size	0.001
	(3.461×10^{-4})
Medicare price	-3.846×10^{-4}
	(2.423×10^{-4})
County FE	Yes

Notes: For each market-year, I calculate the total number of entries ("entry count"). Then I run an ordered logit of entry counts on market characteristics (market fixed effects, market size, Medicare price). I restrict the ordered logit to market-years with 3 or less entries; greater than 3 entries are very rare and give unintuitive results. Estimates of county fixed effects and cutoffs suppressed.

F Hybrid scheme counterfactual with 3 firms

Table 31: Reimbursement schemes that lead to 0.76 visits-per-patient-day by three identical hospices.

Per-day rate	**Per-visit rate**	**Medicare Spending (normalized)**	**Ex-Ante Value Function (normalized)**
197.3	0.0	1.0	1.0
177.0	20.0	0.97	0.93
122.0	80.0	0.92	0.73
61.0	150.0	0.89	0.52

Notes: The rates are in US dollars. The total Medicare spending associated with each scheme is normalized with respect to the spending of the per-day scheme. The average ex-ante value function is normalized with respect to the ex-ante value function of the per-day scheme. As we move down rows, equilibrium quality stays the same yet Medicare spending declines.

G Model estimation and solution

G.1 Demand model

For the demand estimation, I limit my dataset to hospices that enter after 2002 (the earliest period in my data). For these hospices, I see the quality choices over their lifetime, and so can construct the reputation measure accurately for my demand estimation. Please see Appendix A for a more thorough discussion.

For demand estimation, we estimate the following set of equations:

$$ln(s_{jt}) - ln(s_{0t}) = \alpha_{m(j)} + X'_{jt}\beta_d + \sigma_n ln(s_{j|gt}) + \xi_{jt} +$$
$$\eta[\mathrm{a}_{jt} + (1-\tau)\mathrm{a}_{jt-1} + (1-\tau)^2\mathrm{a}_{jt-2} + \ldots]$$

$$\xi_{jt} = \rho\xi_{jt-1} + \varepsilon_{jt}$$

This model is estimated via 2-step nonlinear GMM (the nonlinearity stems from the fact that τ and η enter multiplicatively, and that ξ needs to be differenced out and ρ estimated). Let the matrix of instruments be Z. For the first step, I set the weighting matrix to be the inverse of $Z'Z$. For the second step, I construct the optimal weighting matrix using the residuals from the first-step GMM.

The 2-step GMM method requires estimating a large number of parameters, including county fixed effects and coefficients on hospice characteristics. A trick to easing computational burden is to note that only τ and ρ enter nonlinearly, so I can concentrate out all the linear parameters and minimize over (τ, ρ).

G.2 Supply estimation via Bajari et al. (2007)

G.2.1 How market share is calculated from demand estimates

To calculate profit, I need to calculate market shares of all firms for a given state $\mathbf{x}_m$. This is done using the analytical formulae for nested logit (see Berry (1994)). Let

$$\delta_{jt} = \alpha_m + X'_{jt}\beta_d + \sigma_n ln(s_{j|gt}) + \psi_{jt} + \xi_{jt}$$

where σ_n is the nest parameter that captures correlated preferences within the nest. Then the market share can be written as:

$$s_j(\delta,\sigma_n) = \bar{s}_{j/g}(s,\sigma_n)\bar{d}_g(\delta,\sigma_n) = \frac{e^{\delta/(1-\sigma_n)}}{D_g^{\sigma_n}\left[\sum_g D_g^{(1-\sigma_n)}\right]}$$

where

$$D_g \equiv \sum_{j\in G_g} e^{\delta_j/(1-\sigma_n)}$$

G.2.2 Equilibrium concepts

Markov Perfect Equilibrium: Markov Perfect Equilibrium is a refinement of Nash Equilibrium that is a type of Subgame Perfect Equilibrium. It says that a player's strategy depends only on "payoff-relevant" state variables rather than on potentially the whole history of play. Intuitively, it limits the number of variables that a player looks at before taking an action. This introduces simplicity as well as ruling out a vast number of multiple equilibria. See Ericson and Pakes (1995) for a detailed discussion.

Symmetric and anonymous policy functions: I also impose the policy functions to be symmetric and anonymous. The symmetry assumption imposes that all firms have the same policy function. The anonymity assumption imposes that a firm does not condition its choice on the identity of its rivals.

G.2.3 Forward simulation and optimization

A brief overview of the algorithm is as follows. Fix a guess of structural parameters θ. For every observed state $\mathbf{x}_t$ and hospice j (i.e. each observation in my hospice-level panel data), I estimate the choice-specific value function for each quality level. For quality choice a_j by hospice j, a single simulated path is constructed as follows:

1. The first-stage policy function is used to predict quality choices by all firms, conditional on the simulated market state in that period.
2. The reputation stock of each firm is updated based on its quality choice.
3. Market share of each hospice is realized. Each incumbent receives a payoff.
4. The entry transition function is used to predict the number of potential entrants who apply for entry. These entries are implemented in the next period.
5. The empirical exit probabilities are used to predict whether any incumbent exits.
6. Using the state variable transition functions from first-stage, all state variables update. This includes unobserved demand shocks, market size, and Medicare rates.
7. The game moves to the next period, and I repeat this algorithm.

The above simulation is run for many periods until the discounted profit is driven close to zero; in my estimation I set this to be 80 periods. The simulations use Halton sequences. Summing together the discounted per-period payoffs gives me the total payoff from hospice j choosing quality a_j for a single simulated path. I repeat this simulation 200 times and average over the total payoffs. This gives me the choice-specific value function of hospice j choosing quality a_j in state $\mathbf{x}_t$, $\hat{v}_j\left(a_j, \mathbf{x}_t, \sigma^*; \theta\right)$. In short, I average over the sum of discounted payoffs for each simulation run, and this gives me the choice-specific value function for that action-state combination.

The choice-specific value functions are constructed for every quality $a \in \mathscr{A}$. These are then used to construct model-predicted choice probabilities $\Psi_j\left(a_j, \mathbf{x}_t, \sigma^*; \theta\right)$ from the structural model using Equation (9). The model-predicted choice probabilities are interpreted as follows – given state $\mathbf{x}_t$, how likely is it that hospice j would choose action a_j, conditional on our guess of structural parameters θ.

Next, I describe how these model-predicted choice probabilities are used to estimate the hospice cost function. The intuition is that I search for values of parameters θ that minimize the distance between model-predicted choice probabilities and observed choices.

The dynamic oligopoly model is estimated using two-step GMM. The GMM residual term for observation n is:

$$\Xi_n(\theta) = a_n^{data} - \sum_{a_n \in \mathscr{A}} a_n \hat{\Psi}(a_n, \mathbf{x}_n^{data}, \sigma^*; \theta)$$

where $\hat{\Psi}(a_n, \mathbf{x}_n^{data}, \sigma^*; \theta)$ is the predicted choice probability for action a_n, a_n^{data} is the observed quality choice for observation n, θ is the guess of structural parameters, and $\mathbf{x}_n^{data}$ is the market state for observation n. The resulting GMM optimization problem is:

$$\min_{\theta} \left[\frac{1}{N} \sum_n Z_n' \Xi_n(\theta) \right]' \hat{W} \left[\frac{1}{N} \sum_n Z_n' \Xi_n(\theta) \right]$$

This is solved with 2-step GMM. The instruments used in GMM are the county indicators and the variables in the first-stage empirical policy functions (with the exception of the firm fixed effects), giving us the moment conditions $E[Z'\Xi_n(\theta)]$.

Let the matrix of instruments be Z. For the first step, I set the weighting matrix to be the inverse of $Z'Z$. For the second step, I construct the optimal weighting matrix using the residuals from the first-step GMM.

A way I ease computational burden is by using the "linear-in-parameters" method outlined in Bajari et al. (2007) and Ryan (2012). Similar to their setup, the hospice payoff functions can be split up in such a way that I do not need to redo the forward simulation for each guess of the cost parameters. Instead, I can perform the forward simulation just once and then use the trial cost parameters to rescale the choice-specific value functions and then evaluate the objective function.

Please see Bajari et al. (2007) and Ryan (2012) for a careful explanation.

G.3 Bootstrapping standard errors

Following Wang (2022), I sample blocks of county-year observations from my data with replacement until I match the total number of county-year observations in my dataset. Then I re-estimate the model on each bootstrap sample. I use the sample variance over all the resulting bootstrap estimates to calculate the standard errors for my cost parameters. 400 bootstrap runs are used.

G.4 Counterfactual model solution via policy iteration

G.4.1 Policy iteration

To conduct counterfactuals I solve the model by policy iteration method. This requires solving a nested fixed-point problem; more specifically, I need to calculate $\Psi_j\left(a_j,\mathbf{x}_m,\sigma^*\right)$, $\hat{v}(a_j,\mathbf{x})$ and $V(\mathbf{x})$ such that Equations (14), (15), and (16) simultaneously hold:

$$\Psi_j\left(a_j,\mathbf{x}_m,\sigma^*\right)=\frac{\exp\left\{\hat{v}_j\left(a_j,\mathbf{x}_m,\sigma^*\right)/\theta_e\right\}}{\sum_{a\in\mathscr{A}}\exp\left\{\hat{v}_j\left(a,\mathbf{x}_m,\sigma^*\right)/\theta_e\right\}} \tag{14}$$

$$\hat{v}_j(a_j,\mathbf{x}_m,\sigma^*)=\sum_{\mathbf{a}_{-j}\in\mathscr{A}_{-j}}\left\{\left[\bar{\pi}(a_j,\mathbf{a}_{-j},\mathbf{x}_m)+\beta V(\mathbf{x}'_m,\sigma^*)\right]F(\mathbf{x}'_m|\mathbf{x}_m,a_j,\mathbf{a}_{-j})\prod_{n\neq j}\Psi_n(\mathbf{a}_{-j}[n],\mathbf{x}_m,\sigma^*)\right\} \tag{15}$$

where $\Psi_n(\cdot,\mathbf{x}_m,\sigma^*)$ is the CCP of firm n, $\mathscr{A}_{-j}$ is the set of all possible rival action profile vectors, and $\mathbf{a}_{-j}[n]$ is the action by firm n.

$$V_j(\mathbf{x}_m,\sigma^*) \quad = \quad \theta_e\left[0.577216+\ln\Big(\sum_{a_j\in\mathscr{A}}e^{\hat{v}_j(a_j,\mathbf{x}_m,\sigma^*)/\theta_e}\Big)\right] \tag{16}$$

I solve the model by iterating between the three equations until convergence. Policy iteration proceeds as follows. I iterate between Equations (15) and (16) several times, then iterate between Equations (14) and (15) several times. I repeat this until successive CCPs and ex-ante value functions are within some pre-determined tolerance.

G.5 Absence of multiple equilibria

Estimation and solution of dynamic games are sometimes hampered by presence of multiple equilibria. Fortunately, in my setting, I do not face this issue.

To see if dynamic games in my setting have multiple equilibria, I explored extensively with the dynamic game solution via policy iteration to see if the algorithm ever converges to different equilibria. I did this for heterogeneous firms, varying numbers of firms, different starting points for the policy iteration algorithm, etc. In all the cases I tried, the algorithm converged to the same equilibrium.

In hindsight, this is unsurprising given my model structure (and the institutional setup). Multiple equilibria might have been an issue if firms chose both prices and quality; in such cases, one firm might choose to be low-price low-reputation, while another might choose to be high-price high-reputation. This would lead to multiple equilibria. However, in the hospice industry, firms choose only quality; the price is set by Medicare.

Multiple equilibria are also ruled out under the assumptions made in Appendix G.2.2, which are standard in the dynamic game estimation literature.

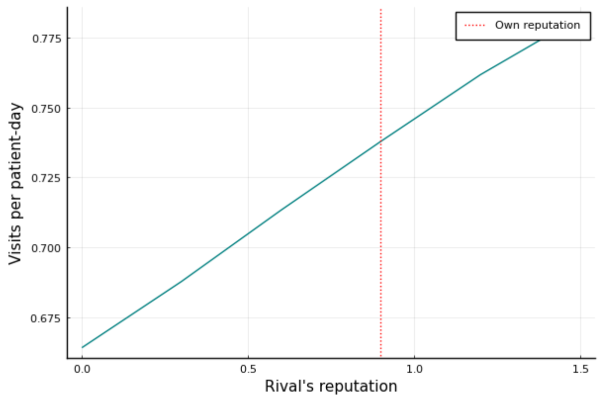
H Illustrating the policy function

In this section, I illustrate the optimal policy functions that I derive in my counterfactuals.

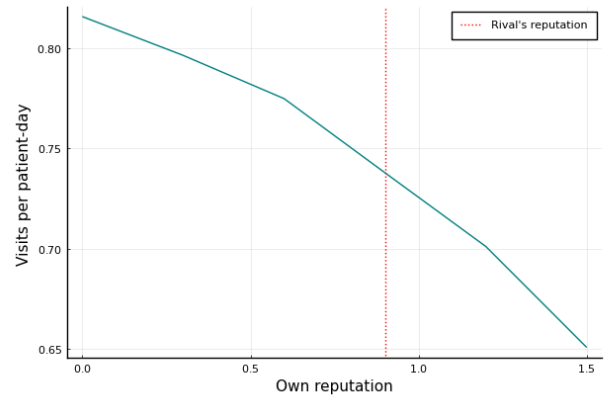
Please not that this is not the empirical policy function used for the cost estimation; for that, see Appendix E.

To illustrate the shape of the policy function, I solve the model for 2 identical hospices and plot the policy function. The y-axis reports the choice-probability-weighted quality choice (i.e.

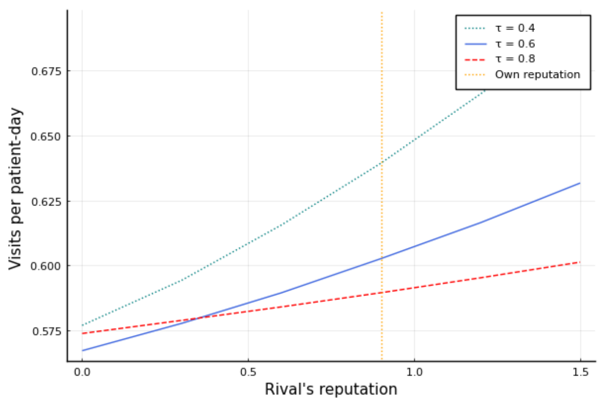
Figure 7: Plotting the policy function against state variables and different τ values.



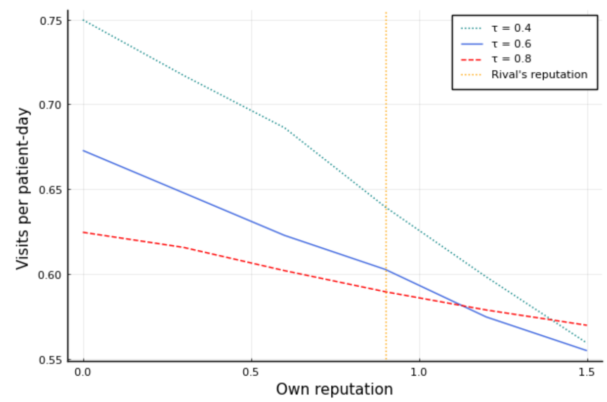
(a) Policy function against rival reputation



(b) Policy function against own reputation



(c) Policy function against rival reputation for different τ



(d) Policy function against own reputation for different τ

Notes: The model is solved for two identical hospices via policy iteration. The y-axis reports $\sum_{a_j \in \mathscr{A}} \Psi(a_j, x_m, \sigma^*) \cdot a_j$, which gives the choice-probability-weighted quality choices. Figures 7a and 7c hold the hospice's own reputation fixed at 0.9. Figures 7b and 7d hold the rival hospice's reputation fixed at 0.9.

$\sum_{a_j \in \mathscr{A}} \Psi(a_j, \cdot, \sigma^*) a_j)$ for the corresponding state in the x-axis.

1. Figure 7a holds reputation of a hospice fixed, and shows that a hospice raises its quality as it faces a rival with higher and higher reputation.

2. Figure 7b holds reputation of the rival hospice fixed, and shows that a hospice lowers quality as its own reputation improves.

3. Figures 7c and 7d show that as reputation depreciates faster (τ gets larger) the policy func-

tions become flatter with respect to own reputation and rival reputation.

I Additional discussion on interpreting the structural model and counterfactuals

I.1 Interpreting entry-exit incentives from ex-ante value functions

I report the ex-ante value functions for my counterfactuals to show how entry-exit incentives change. The overall intuition is that as market conditions improve, potential entrants will choose to enter; as market conditions worsen, incumbents will choose to exit.

I now give two examples to elaborate.

1. In a sequential entry game of homogeneous firms with no private shocks (e.g. Igami (2018)), the equilibrium number of firms n^* is given by:

$$V(n^*, s_0) \geq \kappa_{entry}$$

   where $V(\cdot,\cdot)$ is the value function, s_0 is the starting state, and κ_{entry} is the entry cost. We can see here how a policy change that uniformly raises $V(\cdot,\cdot)$ will raise n^*, and the opposite will lower n^*.

2. In dynamic games studying exit, the exit decision compares the value function from staying on with the scrap value, i.e. $V(x_m) \leq \phi_{scrap}$. If this inequality holds, then subject to logit shocks, the firm is likely to exit. Any policy that lowers $V(\cdot)$ therfore makes exit more likely.

While ex-ante value functions are indicative of how entry-exit incentives change, they do not track them exactly in my setting; this is because there are economic mechanisms I rule out by not explicitly modeling entry and exit decisions. For instance, if entry by new firms is a possibility, then incumbents may react by building up reputation to deter entry. If exit were a possibility, then incumbents will account for the possibility of themselves or their rivals exiting when making quality choice today. I do not allow for these in my counterfactual model solution.

I do not model entry and exit explicitly for several reasons. First, entry is moderate and exit is infrequent in my data. Second, it will increase the computational burden of solving my counterfactuals (both through greater computational need as well as increasing the state space size). Finally, I don't find evidence of entry deterrence through reputation accumulation in my data, and so do not want to include it as an economic mechanism when doing the counterfactuals.

I.2 Interpreting the profit function

Let the profit from patient i be

$$\Pi_i = P_m^{MCAR}\, LOS_i - \gamma_0 - \gamma_1 a_j LOS_i$$

This is a random variable, becaues LOS_i varies by patient i, and is unknown to the hospice when it chooses quality a_j or even during patient enrollment.

The expected profit per patient integrates over the above expression with respect to the distribution of LOS:

$$\mathbb{E}_{LOS}[\Pi_i] = P_m^{MCAR}\, \overline{LOS} - \gamma_0 - \gamma_1 a_j \overline{LOS}$$

where $\mathbb{E}_{LOS}[LOS_i] \equiv \overline{LOS}$ to match the notation in the main text.

Using this notation, the profit function used in the paper is therefore:

$$\bar{\pi}(a_j, \mathbf{a}_{-j}, \mathbf{x}_m; \theta) = -\phi_j + \sum_{i=1}^{M_m s_j(a_j, \mathbf{a}_{-j}, \mathbf{x}_m)} \mathbb{E}_{LOS}[\Pi_i]$$

$$\implies \bar{\pi}(a_j, \mathbf{a}_{-j}, \mathbf{x}_m; \theta) = -\phi_j + \sum_{i=1}^{M_m s_j(a_j, \mathbf{a}_{-j}, \mathbf{x}_m)} P_m^{MCAR}\, \overline{LOS} - \gamma_0 - \gamma_1 a_j \overline{LOS}$$

$$\implies \bar{\pi}(a_j, \mathbf{a}_{-j}, \mathbf{x}_m; \theta) = -\phi_j + M_m s_j(a_j, \mathbf{a}_{-j}, \mathbf{x}_m) \left[P_m^{MCAR} \, \overline{LOS} - \gamma_0 - \gamma_1 a_j \overline{LOS} \right]$$

which is the profit function reported in the main text.

Note that this is different from the profit function being the sum of *realized* profits per patient:

$$\tilde{\pi}(a_j, \mathbf{a}_{-j}, \mathbf{x}_m; \theta) = -\phi_j + \left\{ \sum_{i=1}^{M_m s_j(a_j, \mathbf{a}_{-j}, \mathbf{x}_m)} \Pi_i \right\}$$

The expectation of this profit function lines up with the sum of expected profits:

$$\mathbb{E}_{LOS}[-\phi_j + \left\{ \sum_{i=1}^{M_m s_j(a_j, \mathbf{a}_{-j}, \mathbf{x}_m)} \Pi_i \right\}] = -\phi_j + M_m s_j(a_j, \mathbf{a}_{-j}, \mathbf{x}_m) \left[P_m^{MCAR} \, \overline{LOS} - \gamma_0 - \gamma_1 a_j \overline{LOS} \right]$$

However, it also exhibits variance.

$$\mathbb{V}_{LOS}[-\phi_j + \left\{ \sum_{i=1}^{M_m s_j(a_j, \mathbf{a}_{-j}, \mathbf{x}_m)} \Pi_i \right\}] \neq 0$$

while the sum of expected profits does not

$$\mathbb{V}_{LOS}[-\phi_j + \sum_{i=1}^{M_m s_j(a_j, \mathbf{a}_{-j}, \mathbf{x}_m)} \mathbb{E}_{LOS}[\Pi_i]] = 0$$

J Quotes on hospice reputation

1. Some websites which detail my claim on how hospices are chosen can be found in the following:

   - https://www.vitas.com/hospice-and-palliative-care-basics/when-is-it-time-for-hospice/how-to-choose-a-hospice-provider
   - https://www.cancer.org/treatment/end-of-life-care/hospice-care/how-to-find.html

- https://www.health.harvard.edu/staying-healthy/choosing-hospice
- https://hospicefoundation.org/End-of-Life-Support-and-Resources/Coping-with-Terminal-Illness/How-to-Choose
- https://rainbowhospice.org/hospice-care/choosing-hospice/

2. Some quotes on how reputation should be considered when choosing a hospice are as follows.

   - AmericanHospice.org: What do others say about this hospice? Get references both from people you know and from people in the field – e.g., local hospitals, nursing homes, clinicians. Ask anyone that you have connections to if they have had experience with the hospice and what their impressions are. Geriatric care managers can be a particularly good resource, as they often make referrals to hospices and hear from families about the care that was provided... How long has the hospice been in operation? If it has been around for a while, that's an indication of stability."
   - HospiceFoundation.org: "Seek professional opinions. Ask clinicians, professional caregivers at nursing homes, geriatric care managers, or end-of-life doulas about their experience with a hospice. Talk to friends, family, and neighbors who have used hospice services and get their opinions about the experience with a provider."
   - Vitas.com: "Evaluate the hospice provider's history and reputation before you decide. How long has it been in business? ... What do other patients or families say about their experiences?"
   - Caringinfo.org: "Most hospice programs use family satisfaction surveys to obtain feedback about their services so they can make improvements. Ask the hospice to share a summary of their family satisfaction scores for the last several months with you. You can also ask to see their latest state or Medicare inspection report to see if there are care provision problems. Finally, you could ask to see the hospice provider's list of complaints from the past 12 months."

3. My claim is that visits-per-patient-day is a good proxy for the effort exerted by the hospice, since visits are usually the most costly part of care. Here is a quote from a New York Times article (`https://www.nytimes.com/2023/06/10/health/hospice-profits.html`) from a former hospice admin:

   "In the nearly 20 years that Megan Stainer worked in nursing homes in and around Detroit, she could almost always tell which patients near death were receiving care from nonprofit hospice organizations and which from for-profit hospices."There were really stark differences," said Ms. Stainer, 45, a licensed practical nurse. Looking at their medical charts, "the nonprofit patients always had the most visits: nurses, chaplains, social workers. The nonprofit hospices responded quickly when the nursing home staff requested supplies and equipment."

J.1 Example of a hospice reimbursement scheme

LOCATION BY COUNTIES	ROUTINE HOME CARE DAYS 1-60 Revenue Code 0650	ROUTINE HOME CARE DAYS 61+ Revenue Code 0659	SERVICE INTENSITY ADD-ON (HOURLY) Revenue Code 0552	CONTINUOUS CARE (HOURLY) Revenue Code 0652	INPATIENT RESPITE CARE Revenue Code 0655	GENERAL INPATIENT CARE Revenue Code 0656
NATIONAL RATES	**\$ 196.50**	**\$ 154.41**	**\$ 41.57**	**\$ 41.57**	**\$ 185.27**	**\$ 758.07**
RURAL AREAS	\$ 237.03	\$ 186.26	\$ 50.15	\$ 50.15	\$ 215.38	\$ 903.74
ALAMEDA & CONTRA COSTA	\$ 294.39	\$ 231.33	\$ 62.28	\$ 62.28	\$ 257.98	\$ 1,109.87
BUTTE	\$ 214.74	\$ 168.74	\$ 45.43	\$ 45.43	\$ 198.82	\$ 823.63
FRESNO	\$ 208.62	\$ 163.94	\$ 44.14	\$ 44.14	\$ 194.28	\$ 801.64
IMPERIAL	\$ 180.42	\$ 141.77	\$ 38.17	\$ 38.17	\$ 173.33	\$ 700.28
KERN	\$ 223.41	\$ 175.56	\$ 47.26	\$ 47.26	\$ 205.26	\$ 854.78
KINGS	\$ 206.61	\$ 162.36	\$ 43.71	\$ 43.71	\$ 192.78	\$ 794.41
LOS ANGELES	\$ 234.05	\$ 183.92	\$ 49.52	\$ 49.52	\$ 213.16	\$ 893.02

Figure 8: Snapshot of Medicare reimbursement scheme in 2018. The reimbursement rate varies by county based on its Medicare wage index, and varies by the type of care provided.

K Hospice cost data

Per-period fixed cost:

- Sums: General service cost centers, Other Hospice Service Costs, Bereavement program costs, Other program costs, Writeoffs.
- Subtracts: Revenue from managed care, Revenue from private insurance, Revenue from self-pay, Revenue from other payers, Grants, Donations/Contributions, Unrelated Business Income, Total other operating revenue.

Non-visit cost:

Figure 9: List of variables in the hospice cost data.

DETAIL OF OPERATING EXPENSES
(do not input "$" signs, commas, or decimals; round up to whole dollar)
Use data from Medicare Cost Report where applicable.

Line No.		Total (1)
	General Service Cost Centers	
30	Administrative and General	
	Inpatient Care Service	
31	Inpatient – General Care	
32	Inpatient – Respite Care	
	Program Supervision	
35	Hospice Program / Team Supervision (Non-Visit Wages)	
	Visiting Services	
36	Physician Services	
37	Nursing Care	
38	Rehabilitation Services (PT, OT, Speech)	
39	Medical Social Services – Direct	
40	Spiritual Counseling	
41	Dietary Counseling	
42	Counseling – Other	
43	Home Health Aides and Homemakers	
44	Other Visiting Services	
	Hospice Service Cost Centers	
45	Drugs, Biologicals, and Infusion	
46	Durable Medical Equipment / Oxygen	
47	Patient Transportation	
48	Imaging, Lab, and Diagnostics	
49	Medical Supplies	
50	Outpatient Services (including ER Dept.)	
51	Radiation Therapy	
52	Chemotherapy	
53	Other Hospice Service Costs	
	Other Hospice Costs	
54	Bereavement Program Costs	
55	Volunteer Program Costs	
56	Fundraising Costs	
	Other Costs	
57	Other Program Costs*	
59	Total Operating Expenses	

* Program costs including community education and outreach program costs.

- Sums: Inpatient care service expense (both general and respite care), Radiation therapy cost, Chemotherapy cost, Program supervision cost, Volunteer program costs.
- This is divided by total number of patients in that year to get the non-visit cost per patient ("intercept cost").

Visit cost:

- Drugs/biologicals/infusion cost, Durable medical equipment/oxygen cost, Patient transportation cost, Imaging/lab/diagnostics cost, Medical supplies cost, Outpatient services cost, Visiting services cost (all staff types).
- This is divided by total number of visits in that year to get the cost of an additional visit ("visit cost").

L Heuristic discussion on identification

Intuitively, what is the variation in data that lets me pin down key demand parameters and cost parameters?

In a simplified sense, reputation parameters are identified by regressing current market share on a function of current and past quality choices. The nonlinear regression tells me how much impact past choices have on current share, as well as how important the total reputation stock is for explaining current market share. The nonlinear regression is a more rigorous version of the linear regression shown in Table 5. The reputation stock ψ_{jt} varies separately from ξ_{jt} because it subsumes current and past quality choices by the hospice.

The cost parameters are identified through a revealed preference argument. Let us think about the cost of increasing quality from 0.44 to 0.56. Accounting for the demand system and rivals, an increase in quality will lead to higher market share. So the main reason I would observe a firm choosing 0.44 instead of 0.56 is if the increase in total revenue is more than offset by the corresponding increase in cost.

Table 32: Regression of log of per-period fixed cost, log of intercept cost, and visit cost (from cost data) on county dummies.

	(1)	(2)	(3)
Alameda	13.920***	5.262***	111.949
	(0.111)	(0.157)	(102.670)
Amador	12.306***	6.163***	132.778
	(0.516)	(0.732)	(298.813)
Butte	13.109***	5.609***	118.234
	(0.214)	(0.303)	(167.659)
Contra Costa	13.718***	5.460***	141.194
	(0.133)	(0.188)	(122.592)
Del Norte	10.812***	5.493***	200.617
	(1.264)	(1.792)	(871.181)
El Dorado	13.447***	5.049***	128.679
	(0.298)	(0.422)	(275.492)
Fresno	13.043***	6.028***	114.276
	(0.140)	(0.198)	(129.868)
Humboldt	13.900***	5.157***	119.419
	(0.307)	(0.435)	(298.813)
Imperial	12.814***	7.155***	62.458
	(0.478)	(0.677)	(341.705)
Kern	12.943***	5.717***	370.076***
	(0.132)	(0.187)	(120.234)
Kings	13.280***	5.861***	316.966
	(0.351)	(0.497)	(298.813)
Lake	12.594***	5.407***	110.496
	(0.316)	(0.448)	(308.009)
Madera	9.335***	4.375**	132.881
	(1.264)	(1.792)	(550.983)
Marin	13.708***	6.830***	130.809
	(0.283)	(0.401)	(251.488)
Mariposa	11.003***	6.630***	115.431
	(0.381)	(0.540)	(275.492)
Mendocino	12.982***	6.192***	151.994
	(0.316)	(0.448)	(298.813)
Merced	12.737***	5.164***	100.129
	(0.243)	(0.345)	(202.545)
Monterey	13.732***	6.021***	177.939
	(0.202)	(0.287)	(177.829)
Napa	13.257***	5.133***	116.208
	(0.316)	(0.448)	(298.813)
Nevada	12.922***	6.513***	186.773
	(0.248)	(0.351)	(211.292)
Placer	13.732***	6.263***	160.865
	(0.184)	(0.261)	(177.829)
Sacramento	13.731***	5.434***	128.329
	(0.117)	(0.166)	(107.643)
San Francisco	13.117***	6.368***	200.572
	(0.167)	(0.237)	(145.197)
San Joaquin	12.475***	6.288***	110.236
	(0.174)	(0.246)	(154.004)
San Luis Obispo	13.943***	5.847***	109.273
	(0.217)	(0.307)	(205.339)
San Mateo	14.021***	6.485***	141.582
	(0.156)	(0.221)	(145.197)
Santa Barbara	13.720***	5.944***	108.140
	(0.170)	(0.242)	(155.222)
Santa Clara	14.171***	5.493***	122.940
	(0.124)	(0.176)	(113.902)
Santa Cruz	14.012***	3.892***	139.707
	(0.276)	(0.391)	(246.407)
Shasta	12.472***	4.536***	139.374
	(0.269)	(0.382)	(197.284)
Siskiyou	11.882***	6.034***	1143.690***
	(0.220)	(0.312)	(185.736)
Solano	13.837***	5.298***	173.224
	(0.243)	(0.345)	(232.833)
Sonoma	13.998***	4.554***	145.786
	(0.152)	(0.216)	(126.404)
Stanislaus	13.695***	6.583***	118.777
	(0.193)	(0.273)	(174.236)
Sutter	13.377***	6.824***	139.874
	(0.365)	(0.517)	(262.671)
Tehama	12.896***	2.542***	163.460
	(0.447)	(0.634)	(308.009)
Tulare	13.075***	5.524***	170.015
	(0.264)	(0.374)	(194.802)
Tuolumne	12.959***	5.716***	167.617
	(0.338)	(0.479)	(308.009)
Yolo	13.262***	5.083***	128.558
	(0.316)	(0.448)	(290.394)
Yuba	12.420***	6.142***	206.696
	(0.381)	(0.540)	(355.658)
N	1,495	1,495	1,806
R^2	0.991	0.913	0.038

M Additional discussion on institutional details and modeling decisions

M.1 Advertising

I allow for unobserved demand shocks in my demand model, and advertising gets picked up there. Furthermore, my understanding is that advertising was pretty low in the hospice industry during my data period; in fact, hospices have generally suffered from lack of awareness. I don't model advertising as a strategic choice variable in my supply estimation.

M.2 Reputation index versus perfect information on past quality choices

There is the question of how consumers *know* reputation in my setting. Intuitively, I do not think of individual patients and their families perfectly tracking past quality choices by each hospice. Instead, all consumers need is to be able to observe the reputation index of each hospice. This reputation index can be formed within the community from past experiences with a hospice. The consumer can find out this reputation index by talking to members of the community such as the social worker, physician, support groups, etc. This doesn't require an individual consumer to have carefully kept track of every quality choice made by a hospice in the past.

N Connection to CMS quality measures

In this section, I describe the CMS quality measures, describe and interpret the results of descriptive regressions, report the regression results, then list references for further information.

N.1 Description and discussion of results

CMS tracks 4 broad quality measures for hospices. I describe them below, using references listed in Appendix N.3:

1. The CAHPS Hospice survey is a list of questions given to the primary caregiver of the hospice patient about the hospice. It asks questions regarding: Communication with Family, Getting Timely Help, Treating Patient with Respect, Emotional and Spiritual Support, Help for Pain and Symptoms, Training Family to Care for Patient, Rating of this Hospice, and Willingness to Recommend this Hospice.

   Unfortunately for my purposes, it began in 2018, and furthermore is missing survey data for the majority of hospices; this is because CMS does not report data if a hospice has not received at least 75 survey responses. For the 12/2019 snapshot, only 262 out of nearly 900 hospices in California have scores reported. Furthermore, the results reported seem to be cumulative data – for instance, the 12/2019 snapshot reflects all survey responses received from 1/2017 to 12/2018. So it is not clear how to parse out across-year quality variation either.

   I merge the 12/2019 snapshot of this survey (containing quality data upto 2018) with my 2018 hospice data including the full list of California counties. I use fuzzy-matching on hospice names. Among the observations that I can match, around 150 have reported scores.

   I run some descriptive regressions, with the results reported below. The results are very noisy and rely on small sample size, so they should be interpreted with caution. I find that visits-per-patient-day is positively correlated with the "ratings" by family caregiver for a wide range of specifications; however it is not statistically significant, although the t-stat is around or greater than 1 in most cases. When I regress log of market share on visits-per-patient-day and ratings, I find that visits-per-patient-day is more strongly related to market share than ratings (based on their t-stats).

   When I merge the survey data with my limited subset of 39 counties, there are few observations remaining (around 50). For that smaller dataset, the above regression results do not hold, but it is not clear why.

   Note also that it is not clear review-based quality measures are necessarily better than input-based quality measures, since they are affected by a different set of issues (such as those detailed in Tadelis (2016)).

2. Hospice Visits in Last Days of Life: Proportion of patients who received a visit in last 2 of 3 days of life. This is very similar to my measure of visits-per-patient-day.

3. "HIS Comprehensive Assessment at Admission": This measures the proportion of patients for whom the hospice performed the following 7 care processes – Beliefs/values addressed, Treatment preferences, Pain screening, Pain assessment, Dyspnea treatment, Dyspnea screening, and Treated with an opioid who are given a bowel regimen. To the best of my knowledge, this was first published in 2017 (`https://healthcareprovidersolutions.com/new-hospice-comprehensive-assessment-measure-added-to-hospice-compare/`).

   As far as I can tell, this data is not directly downloadable, so I was not able to work with it. However, one can go into Hospice Compare/Care Compare, click on an individual hospice, and see the *current* score for the individual hospice. I am not aware of how one can find these scores for previous years.

4. "Hospice Care Index": This tracks 10 quality measures – (from the manual) CHC or GIP Provided, Gaps in Skilled Nursing Visits, Early Live Discharges, Late Live Discharges, Burdensome Transitions (Type 1) - Live Discharges from Hospice Followed by Hospitalization and Subsequent Hospice Readmission, Burdensome Transitions (Type 2) - Live Discharges from Hospice Followed by Hospitalization with the Patient Dying in the Hospital, Per-beneficiary Medicare Spending, Skilled Nursing Care Minutes per RHC Day, Skilled Nursing Minutes on Weekend RHC Days, and Visits Near Death. This was first published in 2022.

   From the manual: "Hospices' HCI scores will depend on their summative performance scores on the component indicators included in the index. Each indicator represents a particular care practice of concern, as identified by CMS/s information gathering and measure testing activities. The indicators are assigned a threshold that determines whether that indicator contributes to the hospice/s total index score. These thresholds reflect either normative performance standards (such as providing certain kinds of services) or reference thresholds relative to national hospice performance (such as performing higher or lower than a certain percentage of all hospices). The thresholds have been set based on CMS's statistical analysis of national hospice performance to ensure validity, reliability, and meaningful distinction between hospices... Hospices' index scores are tied to the number of indicator thresholds they meet. Hospices score better if they meet more thresholds for indicator credit and score worse if they meet fewer indicator thresholds. Each indicator for which a hospice meets or exceeds the "Index Earned Point Criterion" adds 1 point to the hospice's overall HCI score. A hospice's HCI score could range from a perfect 10 (if a hospice received credit for all indicator thresholds) to a 0 (if a hospice failed to meet all performance hresholds)."

As far as I can tell, this data is not directly downloadable, so I was not able to work with it. However, one can go into Hospice Compare/Care Compare, click on an individual hospice, and see the *current* score for the individual hospice. I am not aware of how one can find these scores for previous years.

The 2022 document above reports that 85% of the hospices scored 8 or higher out of 10, with 15% hospices scoring lower. It is not clear to me if this HCI is a fine enough measure of quality because of the threshold/indicator nature of measurement. It certainly helps identify the worst-performing hospices.

N.2 "Ratings" data and regression results

The CAHPS data reports "ratings" data from caregivers of the hospices' patients. Each hospice is rated as "9 or 10", "7 or 8", and "6 or lower". For each rating tier, the data reports the percentage of caregivers/family members who rated the hospice in that bracket.

For each hospice, I construct a Hospice Compare Score = (Fraction of caregivers choosing highest tier x 10) + (Fraction of caregivers choosing middle tier x 5) + (Fraction of caregivers choosing lowest tier x 1)

The results are summarized and discussed in the previous subsection.

Table 33: Regression of hospice compare score on visits-per-patient-day and other controls, using all counties in California.

	Hospice Compare Score				
	(1)	(2)	(3)	(4)	(5)
(Intercept)	8.600***	8.681***	8.712***	8.770***	8.128***
	(0.219)	(0.173)	(0.154)	(0.139)	(0.472)
Visits-per-patient-day	-0.062	0.181	0.180	0.200	0.145
	(0.200)	(0.167)	(0.161)	(0.161)	(0.200)
For-profit: 1		-0.405***	-0.364***	-0.325***	-0.215*
		(0.049)	(0.067)	(0.084)	(0.090)
Market size			-0.000		
			(0.000)		
log(total firms)				-0.044	-0.029
				(0.023)	(0.029)
Avg. length of stay					0.002
					(0.003)
Hospice inpatient unit: YES					0.312**
					(0.096)
Pediatric program: YES					-0.189*
					(0.088)
Bereavement for non-hospice survivors: YES					0.198***
					(0.045)
Adult day care: YES					0.136
					(0.378)
Agency type: freestanding					0.338
					(0.333)
Agency type: home-health based					0.072
					(0.426)
Agency type: hospital-based					0.429
					(0.357)
Obs.	157	157	157	157	157

Notes: Standard errors clustered at county-level in parentheses.

Table 34: Regression of market share on hospice compare score and visits-per-patient-day, using all counties in California.

	log(market share)	
	(1)	(2)
(Intercept)	-0.198	0.535
	(1.489)	(1.421)
Visits-per-patient-day	0.500	0.451
	(0.287)	(0.338)
Hospice Compare Score	-0.045	-0.140
	(0.145)	(0.128)
For-profit: 1	-0.467**	-0.324*
	(0.146)	(0.152)
Avg. length of stay	-0.011*	-0.012
	(0.005)	(0.006)
log(total firms)	-0.728***	-0.737***
	(0.047)	(0.052)
Hospice inpatient unit: YES		0.369
		(0.206)
Pediatric program: YES		0.336
		(0.308)
Bereavement for non-hospice survivors: YES		-0.057
		(0.104)
Adult day care: YES		0.399
		(0.494)
Agency type: freestanding		0.193
		(0.249)
Agency type: home-health based		-0.507
		(0.395)
Agency type: hospital-based		0.186
		(0.392)
Obs.	157	157

Notes: Standard errors clustered at county-level in parentheses.

N.3 Links to hospice quality measurement

For additional information, please see:

1. Descriptions of CMS hospice quality measures can be found at: https://www.cms.gov/medicare/quality/hospice/current-measures.

2. The archived snapshots of the datasets can be found at: https://data.cms.gov/provider-data/archived-data/hospice-care.

3. https://www.ahrq.gov/talkingquality/measures/setting/long-term-care/hospice.html

4. https://www.cms.gov/files/document/hospice-care-index-hci-technical-reportjuly-2022.pdf.